A QUICK GUIDE TO

DRY-HARVEST

AGRICULTURAL ENTREPRENEURSHIP







VALUE ADDITION CREATES WEALTH

FROM FARM TO PROSPERITY

The Kings Business Requires a Haste

FATAI O. SALAHUDEEN, PhD

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PROFESSIONAL BOOK REVIEW

A Quick Guide to Dry-Harvest Agricultural Entrepreneurship

Author: Dr. Fatai O. Salahudeen

OVERALL REVIEW

A Quick Guide to Dry-Harvest Agricultural Entrepreneurship is a practical, timely, and highly relevant resource that addresses one of the most critical challenges in agricultural development: the transformation of harvested agricultural produce into sustainable economic value through drying, processing, storage, marketing, and entrepreneurship. The book effectively bridges the gap between agricultural production and agribusiness management by demonstrating how value addition can serve as a catalyst for wealth creation, food security, rural industrialization, and youth empowerment.

One of the most commendable features of the book is its multidisciplinary approach. It integrates concepts from entrepreneurship, agro-processing technology, capacity management, financial planning, risk management, value-chain development, and export logistics into a coherent framework that is both educational and operational. Rather than merely presenting theoretical discussions, the author provides practical tools, case studies, assessment exercises, flashcards, and real-world examples that support experiential learning and business application.

The book is particularly valuable for farmers, agripreneurs, agro-processors, agricultural students, cooperative societies, entrepreneurship trainers, development practitioners, financial institutions, policymakers, and investors seeking opportunities within agricultural value chains. Its emphasis on rural youth empowerment and local enterprise development also makes it a significant contribution to sustainable community development and employment generation.

Overall, this publication successfully transforms agricultural entrepreneurship from an abstract concept into a practical pathway for economic development. It deserves recognition as a useful handbook, training guide, and reference material for agribusiness education and professional development.

COMPARATIVE REVIEW

Compared with many agricultural entrepreneurship texts that focus primarily on production techniques, this book adopts a more comprehensive value-chain perspective. It combines entrepreneurship, technology, management, finance, risk management, logistics, and market development into a single practical guide.

Distinguishing Features

1. Strong focus on dry-harvest value addition.
2. Emphasis on rural youth empowerment and enterprise development.
3. Practical case studies and assessment exercises.
4. Export logistics and market development orientation.
5. Financial risk management and investment analysis.

6. Suitable for certification programmes and entrepreneurship training.
7. Application-focused rather than purely academic.

RECOMMENDATION

This book is highly recommended for:

- Agricultural Entrepreneurs
- Farmer Cooperatives
- Rural Youth Development Programmes
- Agricultural Extension Officers
- Agribusiness Consultants
- Agro-Processors
- Entrepreneurship Development Centres
- TVET Institutions
- Polytechnics and Universities
- ICIME and MCE Professional Certification Programmes
- Development Agencies and NGOs
- Financial Institutions Supporting Agribusiness
- Investors in Agricultural Value Chains

FINAL VERDICT

A Quick Guide to Dry-Harvest Agricultural Entrepreneurship is an outstanding practical handbook that successfully combines agricultural knowledge with entrepreneurship, business management, value-chain development, and rural economic transformation. It provides readers with both the technical and managerial competencies required to establish successful agro-processing enterprises and contributes meaningfully to the broader goals of food security, employment creation, rural industrialization, and sustainable economic development. **Overall Professional Rating: 4.8/5 (Excellent).**

THE AUTHOR

Fatai Oyewole Salahudeen, PhD, hails from Oyo town where he had his primary education from 1971 to 1976.

He later attended All Saint's Anglican Post-Primary Schools, Yaba, Lagos from 1978 to 1982 for his WAEC O' level. He attended Universal Computer Schools, Yaba, Lagos from 1985 to 1987 where he studied Mainframe computer programming and operations. He further had both polytechnic and university education at Yaba College of Technology and Delta State university where he studied statistics and Industrial Mathematics respectively. He obtained B.Sc in Statistics, MBA (O.M) and Doctor of Philosophy in Entrepreneurship from Keisie International University (KIU), USA. He obtained Postgraduate Certificate in Entrepreneurship (the British Institute of Management curriculum) from International School of Management, Lagos.

He is a pioneer graduate of ICAN Accounting Technician Scheme of March 1990 and a Certified Management Consultant (CMC) & Fellow of Institute of Management Consultant (IMC). He holds the Distinction Class of Certified Management Specialist from London Graduate School. He is professional Certified Data Protection Officer with Practice License of the Federal Government of Nigeria under the National Data Protection Act 2023. He is an awardee for Chief Data Officer Ambassador and a Fellow of Institute of Information Management (IIM), He holds the United Nations Distinguished Leadership award in Entrepreneurship.

He has attended several advanced management and industry courses locally and internationally with extensive work life experience spanning 4 decades. He recently became a Platinum Fellow of Chartered Institute of Leadership Manpower and Management Development, USA & Nigeria with full Professional Practice Certificate

Between 1989 and 1992, he had the opportunity to work with several medium scale industries as an accountant and insurance Loss Adjuster consulting finance support and served as a financial and management consultant at Yemi Makinde & Co (Chartered Accountants).

His work-life has been predominantly in the in the finance, manufacturing, and IT sectors of Nigeria, where he has been providing professional services and consultancy on IT/IM for Shell Nigeria and other multinational consulting companies. He was among the consultant staff that developed and managed the Shell Nigeria Computer Center services from 1992 to 1999 and a member of Xerox consultants that developed and managed the Shell Nigeria Electronic Publishing facility from year 2000 to 2002. He is a consultant leading the management of Shell Corporate Publishing and reprographics facility till date.

He has written several published books and unpublished seminar papers, and provides tuition on accounting, computing, entrepreneurship, and management in recent times.

In this study, he brings to bear his wealth and talent of innovative skills and knowledge to farm investments, focusing dry harvest, storage, processing technologies, which could not have come at less time than now.

His great passion includes love for his homeland, country, and family empowerment. He is happily married and blessed with lovable children.

DEDICATION

To all farmers and Entrepreneurs in Nigeria

ACKNOWLEDGEMENT

The successful development of *A Quick Guide to Dry-Harvest Agricultural Entrepreneurship* has been made possible through the inspiration, support, knowledge, and contributions of many individuals and institutions committed to agricultural transformation, entrepreneurship development, and sustainable economic growth.

First and foremost, I express my sincere gratitude to Almighty God for granting the wisdom, strength, and opportunity to complete this work. His guidance and blessings have been the foundation upon which this publication has been accomplished.

I deeply appreciate the countless farmers, agro-processors, entrepreneurs, cooperative societies, agricultural extension practitioners, researchers, and industry professionals whose experiences and practical insights have contributed to the concepts and examples presented in this book. Their dedication to improving agricultural productivity and value addition continues to inspire the development of sustainable agribusiness solutions.

My special appreciation goes to educational institutions, vocational training centres, entrepreneurship development organizations, and agricultural research agencies whose publications, training materials, and field experiences have enriched the knowledge base supporting this guide. Their ongoing efforts in agricultural innovation, food security, technology transfer, and rural development have significantly shaped the practical orientation of this work.

I acknowledge the contributions of development practitioners, policymakers, financial institutions, microfinance organizations, and investment promoters who continue to create opportunities for agricultural enterprises to grow and thrive. Their commitment to supporting rural economies and agribusiness development remains essential to sustainable national development.

A special word of appreciation is extended to trainers, facilitators, consultants, and educators who dedicate their time to building entrepreneurial capacity and preparing the next generation of agribusiness leaders. Their efforts in knowledge transfer and skills development are helping to create self-reliant individuals and prosperous communities.

I am particularly grateful to the many rural youth, aspiring entrepreneurs, women-led enterprises, farmer groups, and small business owners whose determination to succeed under challenging circumstances demonstrates the enormous potential that exists within the agricultural sector. This book is dedicated in part to supporting their aspirations by providing practical guidance for enterprise development, value addition, and wealth creation.

My appreciation also goes to colleagues, associates, reviewers, and well-wishers whose encouragement, constructive suggestions, and professional support contributed to improving the quality and relevance of this publication.

Finally, I acknowledge the invaluable support of the **Institute of Corporate Industrial Management Entrepreneurs (ICIME)** and the **Master College of Entrepreneurship (MCE)** for their commitment to entrepreneurship education, capacity building, professional development, and the promotion of enterprise-driven solutions for economic transformation.

I would also like to express my heartfelt appreciation to the reviewers, subject matter experts, agricultural practitioners, entrepreneurship educators, and professional colleagues who generously devoted their time, expertise, and constructive feedback to the review of this manuscript. Their insightful comments, critical observations, and valuable recommendations have significantly enhanced the quality, accuracy, clarity, and practical relevance of this publication. Their contributions helped to strengthen the technical content, improve the learning structure, and ensure that the book remains useful to students, entrepreneurs, trainers, researchers, policymakers, and agribusiness professionals. I am sincerely grateful for their commitment to excellence and their support in making this work a more comprehensive and impactful resource.

Special thanks are extended to the manuscript reviewers and academic assessors whose rigorous evaluation, professional scrutiny, and thoughtful suggestions contributed to the refinement of the concepts, case studies, assessment materials, and practical applications contained in this book. Their dedication to maintaining high educational and professional standards is deeply appreciated and acknowledged.

It is my sincere hope that this book will contribute meaningfully to agricultural entrepreneurship, rural industrialization, youth empowerment, food security, and sustainable wealth creation for individuals, communities, and nations.

Fatai O. Salahudeen, PhD
President, (ICIME)
Principal, (MCE)
Author

FOREWORD

It is with great pleasure that I present this Foreword to *A Quick Guide to Dry-Harvest Agricultural Entrepreneurship*. This book is a timely and practical contribution to the advancement of agricultural entrepreneurship, agro-industrial development, food security, and sustainable wealth creation. At a time when post-harvest losses continue to limit the economic potential of agriculture, this publication provides a clear roadmap for transforming agricultural produce into profitable products, viable enterprises, and sustainable industries.

One of the greatest challenges facing agricultural development, particularly in emerging economies, is the inability to effectively preserve, process, store, and market harvested products. As a result, valuable crops are lost, incomes are reduced, and economic opportunities remain untapped. The dry-harvest approach presented in this book offers a practical solution to these challenges by emphasizing value addition, appropriate technology, entrepreneurship, quality management, market development, and export readiness.

The strength of this guide lies in its comprehensive treatment of the entire agricultural value chain. It moves beyond cultivation and harvesting to address critical areas such as drying technologies, capacity management, agro-processing, storage systems, packaging, logistics, distribution, financial management, export opportunities, and risk management. The book combines technical knowledge with entrepreneurial thinking, enabling readers to understand not only how products are processed but also how profitable and sustainable agribusinesses are built and managed.

This publication is specifically designed for a wide range of stakeholders, including:

- Agricultural entrepreneurs and agribusiness start-ups
- Farmers, farmer cooperatives, and producer associations
- Agro-processors and food manufacturing enterprises
- Small and medium-sized enterprises (SMEs) in the agricultural sector
- Agricultural extension officers and rural development practitioners
- Students of agriculture, agribusiness, entrepreneurship, and management
- Polytechnics, colleges of agriculture, universities, and vocational institutions
- Participants in entrepreneurship development and skills acquisition programmes
- Financial institutions, microfinance providers, and agricultural investment agencies
- Development partners, NGOs, and community development organizations
- Government agencies responsible for agriculture, commerce, industry, and rural development
- Export promotion organizations and international trade facilitators
- Consultants, trainers, and facilitators in agribusiness and enterprise development
- Professional certification candidates in entrepreneurship, agribusiness, and agro-industrial management
- Investors seeking opportunities within agricultural value chains and food-processing industries

The inclusion of practical exercises, case studies, flashcards, assessment questions, financial analyses, export logistics scenarios, and business planning tools further enhances the usefulness of this book as both a learning resource and a practical field guide. Readers are equipped not only with theoretical concepts but also with the

competencies required to establish, expand, and sustain successful agro-processing ventures.

The author's extensive experience in process management improvement, entrepreneurship development, organizational capacity building, and business transformation is evident throughout the text. The material is presented in a simple, structured, and application-focused manner that makes it accessible to both beginners and experienced practitioners alike.

As nations increasingly seek sustainable pathways to economic diversification, employment creation, food security, import substitution, and export growth, agricultural entrepreneurship will continue to play a strategic role. The future of agriculture lies not merely in production but in processing, innovation, value addition, and market expansion. This book provides a practical framework for achieving these objectives and empowering individuals, communities, and institutions to unlock the full potential of agricultural resources.

A particularly important contribution of this book is its focus on empowering rural youth through entrepreneurship, skills development, and enterprise creation. Across many rural communities, young people face significant challenges including unemployment, underemployment, rural-to-urban migration, and limited access to sustainable income opportunities. This book demonstrates how agricultural value addition, dry-harvest processing, agro-industrial production, marketing, and export development can become viable pathways for economic empowerment. By providing practical knowledge, business tools, case studies, financial planning techniques, and enterprise development strategies, the book equips rural youth with the competencies needed to establish profitable agribusiness ventures, create employment opportunities, build local industries, and become active contributors to community development. It encourages a shift from viewing agriculture merely as subsistence farming to recognizing it as a modern, technology-driven enterprise capable of generating wealth, innovation, and long-term economic prosperity.

Furthermore, this publication serves as a practical resource for youth empowerment programmes, agricultural incubation centres, technical and vocational education institutions, entrepreneurship development agencies, cooperative societies, and rural development initiatives. By fostering entrepreneurial thinking and promoting value-chain participation, the book contributes to the development of a new generation of rural agripreneurs capable of transforming agricultural resources into sustainable businesses, creating jobs, reducing poverty, and strengthening national food security.

We therefore wholeheartedly recommend this publication to entrepreneurs, students, educators, policymakers, development practitioners, investors, cooperative societies, and all stakeholders committed to promoting agricultural transformation and inclusive economic development. I am confident that this guide will become a valuable reference for learning, teaching, enterprise development, and professional practice in the agricultural sector.

Education Board, (ICIME)

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CHAPTER ONE

DRY HARVEST ENTERPRENUERSHIP

1.1 INTRODUCTION

Dry Harvest Management refers to the collection of technology systems and activities for the processing of intermediate food products in agricultural value-chain management.

In the first place, the factory operations tend to provide cash flow to farmers to finance their farm production just as it enables the urban and rural consumers and industrial users to have access to intermediate or semi-finished farm products.

The central objective of the concept of Dry Harvest Production Management, therefore, is the backward deployment and integration of technologies for the abundant wealth in agriculture between the rural and urban economies. This builds capacity and empowerment that foster industrialization in food production. This is one solution that can sustain moving the national and local economies to self-sufficiency and self-reliant from the grassroots unstoppably.

1.2 MONEY AND ENTERPRENUER

In Nigeria, most enterprises are usually informal and self-employed with fewer workers requiring micro-credit to operate as sole proprietors without formal registration. Most are regarded as business undertakings of the poor. Incidentally, the poor have become targets in development programmes, because we can learn from money-managing efforts of the poor and use that learning to design new and better ways of bringing capacity management services to the

spots where micro enterprises strive and succeed. A poor person in Nigeria is described as a person earning below the national minimum wage or not more than twice of per capital income, whichever is higher (CBN). Hence the level of poverty of the poor is highly influenced by the national economic management.

1.3 THE MICRO ENTERPRISE

Self-employment requires assets in the form of human capital (education, experience, and entrepreneurship), financial capital (loans or savings), and social capital (networks and personal support). Proper Micro enterprise capacity building and management would help people of modest means to start, strengthen or expand very small companies. Self-employment requires a wide range of skills, attitudes, and habits. The people most likely to succeed are the elderly educated with experience in self-employment or in wage jobs in the same field, who are cash and social socially prudent.

1.4 TRAINING AND PLANNING CAPACITY

Training is very vital in sustainable micro factory asset building. Training aims to equip micro enterprise owners with the human capital needed to handle the technology and compete in the marketplace through:

1. Advice about the risks of micro enterprises and alternatives.
2. Help with business plans.
3. General skills in accounting, planning, and pricing
4. Assistance with legal compliance and licensing for specific businesses.

5. Consulting services and technical assistance on an on-going, on-call basis.

A critical element of self-employment training is to help clients understand whether they are suited as entrepreneurs. Good training helps the poor to assess their assets, get more education, and to search for skills in vocational jobs, and then attempt self-employment. For one thing, planning forces owners to ask some tough questions:

1. Does the idea make sense in business terms?
2. How long will it take to break even, and how will I get by then?
3. Have I confused the enjoyment of supply with the existence of demand?
4. If I make it, will they come? That is, customers being the king.

Most owners start making a product because they like it rather than because it has a profitable market. Business plan greatly serves three good purposes:

1. For micro enterprise management that is based on loans, the business plan screens out the highest-risk borrowers - no plan, no loan.
2. For management that does not make loans, the business plan prepares borrowers to approach banks or other lenders.
3. Finally, the plan serves as a “final exam” and to track effort toward self-employment.

Sector-specific training makes sense in some fields such as food service that not only have many micro enterprises -

making it worthwhile for the micro enterprise management providers to acquire sector-specific expertise.

1.5 MICROFINANCE

Today, microfinance seems to be the provision of small loans to the poor. Products (loans) and the market (the poor) fall within the purview of microfinance, but they are more its origins than its present and future. The microfinance industry has developed from a history of micro credit programmes. Today, it has grown to cover a broader range of products and services, from credit and savings to insurance and money transfers. Today, the general term of microfinance is the provision of financial services to those excluded from the formal financial system. The services noted here hint at the potential breadth of 'micro financial' services. Reaching "those excluded from formal financial services" means an expanding focus for microfinance. Systems of exclusion are based not just on lack of wealth, but also on social, cultural and gender barriers to financial services. Effective microfinance is positioned to overcome a variety of access barriers to a wide range of financial services for the many different customers who are excluded from the formal financial system.

The future of micro and small businesses is the future of micro finance. Recent research highlights the various levels of poverty of the entrepreneurial groups reached by microfinance services. This research recognizes that poor and low-income people slip from one poverty category to another as opportunities and risks change. This is because microfinance has come to the rescue of many micro investors

by granting loans without the usual stringent requirements of universal banks. The micro financial institutions have proved to be a bailout to revive ailing micro sector particularly in Nigeria. Many have had their entrepreneurial skills upgraded and are now better able to manage their investment better. However, a shift from the concept of 'credit need' to 'repayment capacity' limits risk to MFIs and lowers unbearable debt burdens for their poor clients.

1.6 MICROLOANS

Lack of access to capital is the greatest single barrier to starting businesses, and capital is an input even more important than general business training. There is a strong link between business start-up and owner's net worth. Banks require signals of creditworthiness that micro enterprises often cannot offer real estate for collateral, an unblemished credit record, and a wage job with enough steady income to make monthly payments. Unlike banks, micro enterprise management often uses joint-liability groups to control risk. Managers that lend to individuals also do consider non-standard signals of creditworthiness that banks ignore, such as a record of on-time payments for rent and utilities, regular remittances to relatives abroad, and/or completion of a micro enterprise training course or business plan. A few banks work in partnership with micro enterprise managers to make loans to unemployed "graduates" based on their business plans. This goes to confirm that micro loans are good business. However, a "score point" for the SMEDAN Act in Nigeria, banks until recent times does not seem to have found a profitable niche in micro enterprises. In fact, rather than making loans themselves, some banks score Credit Risk

Assessment points by donating funds for micro enterprise managers to lend.

1.7 MICROSAVINGS

The word “microfinance” did not exist for too long in Nigeria. Conventional banks only spoke of “micro credit”. The new shift through government policy and empowerment programmes shows the awareness that, for micro enterprises, savings matter much if not more as loans. Most new ventures do not use much debt at start-up. In the Nigeria nascent democratic social and economic development, start-ups financial capital most common sources are not bank loans but rather owner’s savings or informal loans from friends or relatives. Formal loans are little used in financing business start-ups. Micro enterprises are limited less by low debt than by low savings because lenders require collateral, low savings lead to low access to loans. Savings have several advantages over loans. Savings pay interest, but loans cost interest, and loans are tied to specific purchases and cannot pay for research during the planning stage or during the years of gestation required of most small ventures. A micro enterprise invests a lot of “pre-venture capital” long before it gets in the sights of a bank. Even if established, successful micro enterprises do not provide monthly payment and need savings on income and expenses. Without a reserve of savings, a new small business will fold before the start.

1.8 SOCIAL SUPPORT

In addition to human capital and financial capital, self-employment requires social capital. Business happens not only in the marketplace but also in the social arena, and the

self-employed must rely on their personal networks for contacts and information and on their families for emotional support and physical sustenance. Self-employment is difficult for anyone, and micro enterprises are not a panacea for disadvantaged groups.

“Business failure and self-employment exit rates are higher among people who choose self-employment without appropriate education, skills, and financial resources. In the long term, the best way to promote self-employment for disadvantaged groups is through improved educational opportunities. Self-employment is not the same as entrepreneurship, and many people of modest means enter micro enterprise not because they are risk-takers who want to get rich through “creativity” but because they cannot find a good wage job and so need a stop-gap measure as they tread until something better comes along.

To increase the chances that slow-growing and or no-growing will transform into the fast-growing and subsequent micro economic growth in Nigeria. Micro enterprise support must offer “self-esteem training” to buttress resilience that inspire “entrepreneurial spirits. Even for motivated low-income, micro enterprise training does not generally increase living standards in the short run although it does make owners feel better. Overall, micro enterprises greatly benefit society at large, by creating jobs to help the poor people not to be over dependent on public assistance and burden of inadequacy.

1.9 BUSINESS EFFICIENCY MEASUREMENT

Business opportunities are on the increase in the different sectors of the agriculture and food technology in Nigeria, but no proper study to make a sound investment decision. Productivity measures are identified as six different categories and are very key in this regard:

1. General Economy
2. Operational Supervision – management availability
3. Staff Relations or human resources available
4. Service Conditions – sectorial attractions
5. Physical Facilities or Infrastructure
6. Capital funding

Efficiency means the optimal level of interplaying resources to meet the needs for achieving a goal. The five classes of efficiency estimation usually adopted are:

1. Very low	10 – 40%
2. Low	41- 60 %
3. Average	61 – 80 %
4. Very Good	81 – 90 %
5. Excellent	91 – 100%

1.10 FLASHCARDS- DRY HARVEST ENTREPRENEURSHIP

Q: What is Dry Harvest Management?

A: The management of harvesting, drying, storage, processing, packaging, distribution, and marketing of agricultural products to increase value and profitability.

Q: What are the three major forms of capital for entrepreneurs?

A:

- Human Capital
- Financial Capital
- Social Capital

Q: What is Human Capital?

A: Knowledge, skills, experience, and competence possessed by individuals.

Q: What is Social Capital?

A: Networks, relationships, partnerships, and community support that facilitate business success.

Q: Why is a business plan important?

A: It guides operations, supports financing decisions, identifies risks, and improves business sustainability.

Q: What is Microfinance?

A: Financial services provided to low-income entrepreneurs who lack access to traditional banking services.

Multiple Choice Questions (MCQs)

1. Dry Harvest Management primarily focuses on:

a) Crop cultivation

b) Livestock breeding
c) Processing intermediate food products
d) Export documentation
Answer: c

2. Which of the following is a key objective of Dry Harvest Production Management?
a) Import substitution only
b) Agricultural taxation
c) Rural-urban integration through agro-processing
d) Land acquisition
Answer: c

3. Micro enterprises generally require:
a) Large foreign investment
b) Human, financial, and social capital
c) Government ownership
d) Import licenses
Answer: b

4. Business plans help entrepreneurs by:
a) Avoiding legal requirements
b) Screening investment risks
c) Eliminating competition
d) Removing need for financing
Answer: b

5. Microfinance provides:
a) Agricultural insurance only
b) Financial services for excluded groups
c) Tax exemptions
d) Free grants
Answer: b

Short Answer Questions

1. Define Dry Harvest Management.
2. Explain the role of microfinance in agricultural entrepreneurship.
3. Describe three forms of capital needed for self-employment.
4. What are the benefits of a business plan?
5. Explain the concept of social capital.

Essay Questions

1. Discuss the importance of entrepreneurship training in sustainable agro-processing business development.
2. Explain how micro enterprises can reduce rural poverty and promote economic growth.
3. Evaluate the role of business planning in Dry Harvest Entrepreneurship.

Practice Case Study

A young entrepreneur wants to establish a cassava drying business in Oyo State with limited capital.

Tasks:

1. Identify required human, financial, and social capital.
2. Prepare a simple business startup plan.
3. Recommend microfinance options.
4. Identify major risks and mitigation measures.

CHAPTER TWO

CAPACITY MANAGEMENT CONCEPT

2.1 PRODUCTION CONCEPT

With regards to agro processing factory, post-harvest capacity is a combination of the storage of raw items, the dryer warehouse and drying platform that complement the scalable labour and machine hours.

2.2 PROFITABILITY CONCEPT

The focus for establishing a farm gate business must be profit making from the operations and management of intermediate food processing. The modality to achieve this aim is to create added economic and commercial values to perishable crops through the process of drying and hygienic storage that increases the shelf-life span of the products for consumption, further industrial and export use.

The commonly identified products are raw cassava, yam, plantain, banana, fish, and potatoes, which are believed to have high significant commercial values when in dried state. These crops are naturally available from farm gates and consumer market across Nigeria. Farm gate technology combines production systems that gradually transform the labor-intensive methods to achieve a higher output and quality method. The system features technical components such as:

1. Sizeable units of concrete-drying platform
2. Sizeable Single loading heat room.
3. Sizeable Raw Products loading store
4. Sizeable, finished products store

5. Motorized chipping machines.
6. motorized milling machine
7. Sizeable washing bay
8. Sizeable product handling bay
9. Sizeable machine room
10. Alternate heat energy system

The manpower source is mostly 100% local skills content and normally abundant in the local environment where the factory can be situated. The factory operation is expected to provide a high return on investment as it relies basically on 100% local input products availability and innovated mechanized technology. The revenue potential is high because there is demand for the finished and semi-finished products from local consumption, industrial usage, and export. The total initial investment is scalable but capable of a conservative ROI of 20%, which is also reinvestable for expansion. This gives a payback and exit period of approximately 5 years.

2.3 OPERATIONS CONCEPT

Production:

1. Farm gate procurement
2. Process products into intermediate products (chips)
3. Process products into final product (flour).

Processing:

1. Locate plant close to primary growing area
2. Limit size of plant to small scale
3. Diversify processing among products lines (yam, plantain, potatoes, fish, etc)

4. Develop contractual arrangements with industrial buyers, resellers, and exporters

Trading:

1. Open retail sales outlet within factory for higher margin to recover losses
2. Sell high quantities in cash with sales discounts.
3. Secure sale contract before buying roots
4. Package products into small sizes for retail sales

2.4 INDUSTRY

The food milling industry in Nigeria is presently undergoing radical research and development by the day as all tiers of government and agencies are getting mandatorily involved including all sectors of the banking industry. Principally, institutions such as IITA, FIIRO, PRODA, and RMDC are all champions in the development of full agricultural input products and processing technologies. This is owing to the local abundant availability of products in the first place, and high potentials of Nigeria land and environment for sustainable agricultural production. This business plan considers networking advantages with primary producers, intermediate processors, consumer, and industrial users in Nigeria.

2.5 CORE ACTIVITIES

The short run activities are to produce semi-finished food products such as: Chips, feedstock, cassava Flour, Gari, Lafun, Poundo Yam flour, Elubo, Plantain Flour, dried fish, potato flour while The long-run production activities involve the production of Confectioneries, Bread, Starch, Ethanol, training and consultancy etc.

Product and/or service description

1. *Cassava products:* Dried cassava root, it is used to produce flour to replace wheat flour and is so used by some people with allergies to other grain crops.
2. *Plantain/Banana Products:* Dried plantains produce gluten-free flour, which can be used in bread, cakes, cookies, and rolls. Plantain and its associated products are useful food items for diabetics.
3. *Yam products:* Dried yam used to prepare the brownish/whitish pastry meal called amala or pounded yam eaten with stew or soup as prestigious meals available in African restaurants. This African food items are suitable for vegetarians and non-vegetarians alike.
4. *Fish Products:* Dried fish produce long shelf-life source of protein for meals and health foods.
5. *Potato products:* Dried potato tubers prepared to produce flours rich in carbohydrate and several energy foods as confectioneries, biscuits, cake, and bread

Customers and marketing-chain

This includes dry food users and consumers, catering services, fast foods, bakeries, millers, pharmaceutical industries, and starch production industries. Cassava and plantain flour have both been recently accepted as alternatives to grain wheat products.

Suppliers and supply-chain

This includes the cassava, yam, banana, plantain, and potato growers as well as fish farmers all over the agricultural belt of Nigeria. Most farmers are primarily growers, and this is the major traditional trade of the communities South-West

of Nigeria. The research and technology development by RMRD, PRODA, FIIRO, IITA, have greatly enhanced the growth, capability and stability of the primary producers and production of agricultural products with modern farming techniques and quality assurance. Production is non-seasonal, and supply is available throughout the year. Prices are open- market and farm-gate based.

Opportunities and weaknesses

i. The business opportunities span multiple customers, which include households, industrial users, and export. Specifically, they are Flour Milling Industries, Bakers and Confectioners, Pharmaceutical industries, Brewers, Livestock industries, Hospitalities, exporters, individuals, and families with annual income from below and above N600, 000, and between the ages of 4 to 70 years. The product geographic catchments for sales and marketing spans the urban cities, and state capitals nationwide. Export markets include EU, UK, US, Brazil, China, Russia, Africa, countries.

ii. The choices of opportunity are based on Consumer population, proximity, industrial spread, transportation network, communication, urbanization, commercial activities, and local content technology. The potential customer based for this study therefore is 100 with spending ability on products above 60% of national minimum wage of N18, 000.00 (i.e N10, 500.00) monthly.

The known direct and indirect competitors include wheat and grain food processors, Dry food importers and emerging processors alike, which include large scale processors, and

exporters with higher industrial capacities. There is no known direct competition with exporters and already established processors in the country as most are located within the urban cities across the country and have superior financial production and management capabilities with track records of successful operations. This is the reason the study looks at a direct marketing strategy and intermediate processing that feeds the industrial competitors as identified to penetrate and establish market catchment for all the product lines.

i. The strengths and weaknesses of some segments of the market are that the retailers are strong in the consumer market but weak in the bulk purchases. The wholesalers have stronger link between the retailers and producers and may frustrate the distribution channels by forming cartels. But may be weakened by the quality and taste of consumers and industrial producers, the prevailing or emerging industrial and economic policies of the governments to protect the infant investors, as well as the product pricing.

ii. The competitive advantages in the operations are that this category of agricultural manufacturing presently does not exist in most towns in Nigeria. The composite flour production is also an emerging industrial sector in Nigerian that needs champions for growth. The present cottage and micro level use crude methods for subsistence production only. The design study mode of operations proposes an intermediate processing technology serving both the primary and industrial producers of the food and manufacturing industries. Mechanization of the production

method will produce high-quality products that are in high demand. This is not as achievable at present. The various government initiatives also empower entrepreneurs in this sector to do better competitively.

iii. Objectively, the strengths of the investment are strong since the owners are prepared to make a success of the agricultural manufacturing. There is need for adequate experience in the management of production, operations, accounting, economics, and information technology systems for the business. The start-up scale is to grow from micro to small scale level, which should be within the management capabilities and experiences of the owners. The sub sector also promises easy access for new entrants as Nigeria presently requires over 200 small-scale plants with about 1000t per annum to meet up with the demands. The present gap in the production level and export guarantees a successful growth to entrants into the sector and markets. The products manufactured are in high demands both in the rural and more in the urban population who can afford the prices. Continuous demand means that the business can be self-sustaining and attain early break-even.

iv. Conversely the Weaknesses may include ability to finance the operation in the short run, which may be depended on the owners' personal savings. This is likely to impact on expansion and ability to meet up with surges in demand especially by the industrial buyers. Another major weakness is the ability to finance research and development, and sourcing for the mechanized technologies. This is where

financial partners are sought for to improve working capital that guarantees uninterrupted profitable operations.

2.6 MANAGEMENT CONCEPT

There is strong cynicism that the "intermediate food production fever" will not last. But this is contrary to the historical and evolutional development of the dry stuff potentials of cassava; yam, plantain, potatoes, fish, and banana for economic uses just to mention a few. Sustaining the potential has brought about a great demand for intermediate processing champions. The factories at the farm gate are thus an invitation to invest and manage or to join others to fully earnest the great potential yet untapped in this rural industrial sub sector. In the long run, the management system becomes self-sustaining and viable for reinvested.

2.7 SHORT-LONG RUN CONCEPT

The primary operation is to include trading and intermediate processing of food products from cassava, yam, plantain, banana, potatoes, fish, etc and any other perishable crops for export, industrial and domestic consumption in the short and long runs. The short run activities involve storage services, processing chips, pellets, and grits. The long run is to involve production of flour for confectioneries, baking, starch, and ethanol.

2.8 INDUSTRIAL-CHAIN CONCEPT

The factory is targeted at generating industrial activities as a chain in the intermediate processing sector that serve as industrial feeders providing inputs for wider uses and export by:

a. Partnering with farmers in the rural areas to establish self-sustaining intermediate processing technologies for farm products for wider industrial applications.
b. Operating mechanized storage systems to achieve elongated lifespan for intermediate food products.
c. Dealing in semi or finished food products for industrial uses and exports.

2.9 NATIONAL CAPACITY CONCEPT

According to studies, the Nigerian flour milling industry requires about 200,000mt of composite flour per annum. This will require 200 plants of 1000t capacity per year for composite flour alone. This is the basis of the factory investment focus at a minimum. A gradual growth to scalable capacity is achievable over years. It is thus strongly believed that there is a wide capacity window with local inputs in Nigeria to achieve the systemic and sustainable growth from 1000mt to 10,000mt per annum.

2.10 INDUSTRY CHALLENGES CONCEPT

a. Food crops are mostly grown by farmers in a rotational cycle.
b. Production is on small micro scales.
c. food processing ventures are localized with manual tools and technologies,
d. there are the difficulties of sufficient supply to the intermediate production,
e. Most crops are highly perishable and must be processed within hours after harvesting.

2.11 PRODUCTION CHALLENGES CONCEPT

The industry challenges pose technological and managerial operation such as:

a. Skilled production and management staff
b. Uninterrupted Power supply
c. Plant and equipment running cost
d. Procurement logistics
e. Procurement financing
f. Transportation financing
g. Sales and marketing

2.12 PRODUCTION STRATEGIES CONCEPT

a. Organized Primary Growers Network (OPGN), for regular product supply.
b. Self-managed Micro Capacity Credit Systems (MCCS), where participants build capacity.
c. Primary Distributed Storage Service (PDSS), where fresh products are preserved at locations within the mobility range of growers.
d. First-level Product Preparation Service (FPPS), where quality and quantity of product are determined.

These strategies generally resolve the traditional problems related to perishable product processing, and:

a. Ensures value-added refining and packaging.
b. Stabilizes the quality and quantity of supply
c. Sustains continuous product input/output all year round.

d. Establishes a self-managed *credit scheme*, ensures income security of suppliers and growers in the rural economy.

2.13 PRODUCTION SCALES

The processing operations are generally at 5 levels of capacity identified as Household (or cottage), Micro, Small, Medium and Large.

Household processing typically does not employ any outside labour. The household consumes virtually all the processed and sells a small amount to raise income for household needs. At present, most Nigerian processors fall within this category.

Micro processing employs one or two units of labour at minimum. This enterprise typically uses batch processing. Batch processing may take four hours per day, and this would be sufficient for the owner. Nigeria has a few or no organized processors in this category of operation.

The small and medium processing typically employ three to ten workers and are very sparse at present. As planned, Dareslauf capacity build-up is calculated on 100% local product supply, and machinery including energy and manpower.

The large-scale processing is not yet in Nigeria are at best very scanty with below capacity operations. Large-scale operations are defined as enterprises employing 10-30 or more labours with large tonnage storage and marketing. The main production line and capacities expected are:

i. Dried Chips
ii. Dried Flour
iii. Dried fish

2.14 SCALLABLE TECHNOLOGY CONCEPT

1. Plant and Machinery
Motorized grating, Milling, Pelleting, and chipping r diesel engine Machines
Weighing scale 1-50kg capacity and Hydraulic Jack Press
2. Factory Building
Factory Building and land, Dryer-Warehouse, and land
Drying concrete platform, Water supply system
Heating equipment/accessories, Trays, and packaging
3. Transport Facilities
Motor Vehicle, and or Motorcycle
5. Pre-operational Expenses
Installation Expenses
Management, permit and Licensing

2.15 FLASCARDS - CAPACITY MANAGEMENT

Q: What is Capacity Management?

A: Planning, controlling, and optimizing resources to meet production targets efficiently.

Q: What is Capacity Utilization?

A: The percentage of total production capacity currently being used.

Q: What is ROI?

A: Return on Investment. A measure used to evaluate the profitability of an investment.
Formula: ROI = Profit ÷ Investment × 100

Q: What is the Payback Period?

A: The time required to recover the initial investment from business profits.

Q: Name three institutions supporting agro-processing in Nigeria.

A:

- FIIRO
- IITA
- PRODA

Q: What is the Industrial Chain Concept?

A: Linking production, processing, storage, distribution, marketing, and exports into one value system.

Multiple Choice Questions

Capacity management refers to:

a) Product branding

b) Managing production resources efficiently

c) Tax administration
d) Warehousing alone
Answer: b

1. The projected ROI of the Dry Harvest Factory is approximately:

a) 5%
b) 10%
c) 20%
d) 50%
Answer: c

2. The projected payback period is:

a) 2 years
b) 3 years
c) 5 years
d) 10 years
Answer: c

3. Which institution supports agro-processing technology development?

a) IITA
b) FIIRO
c) PRODA
d) All of the above
Answer: d

4. Production scales include:

a) Household
b) Micro
c) Small/Medium
d) All of the above
Answer: d

Short Answer Questions

1. Explain the profitability concept in dry-harvest operations.
2. Describe the operations concept.
3. What is the industrial-chain concept?
4. Explain the national capacity concept.
5. Identify five industry challenges.

Practice Exercise

Develop a capacity plan for a 1,000-ton annual cassava flour processing facility.

Include:

- Storage requirements
- Production facilities
- Human resources
- Financing requirements
- Revenue projections

Essay Questions

1. Discuss the strengths and weaknesses of Dry Harvest agro-processing investments.
2. Explain the production strategies required for successful capacity management.

CHAPTER THREE

TECHNOLOGY DEVELOPMENT

3.1 Primary NETWORKING STRATEGIES

This is based on the knowledge of primary sources of raw crops, labour, and strategic locations. The Primary Networking System (PNS) is a necessary part of the development set up. This concept includes a Distributed Storage Service (DSS) whereby the fresh products are preserved at locations within the mobility range of growers. The main purpose of the DSS is simply to keep the freshness of the products before the intermediate processing to sustain the expected high-quality shelf-life**.**

The storage processing concept resolves the traditional problem related to perishable product processing. The objective is to stabilize the quality and quantity of supply to the factory all the year round to sustain continuous output. The storage service assures income security for suppliers, as well as redistributing income into the rural economy. The Networking strategy is a potential background for a managed *credit scheme for micro capacity building.*

3.2 DRYING PRODUCTION SYSTEM

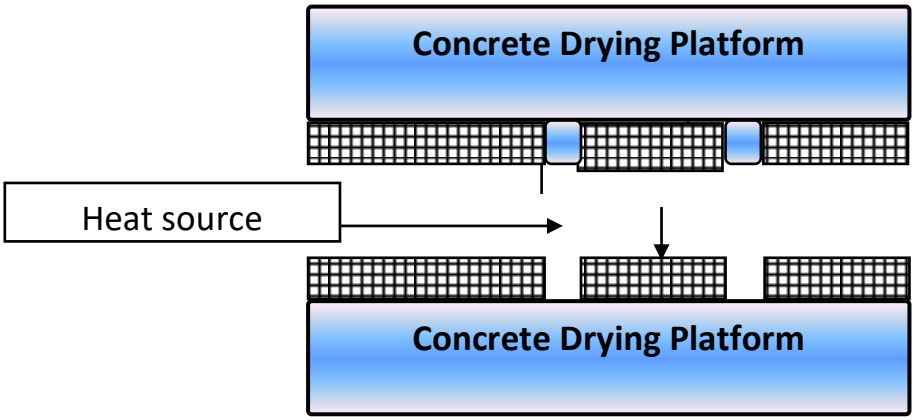
The drying production is the nerve of the dry harvest processing system. Without a robust drying system, the quality and quantity of output products cannot be guaranteed. The energy supply system is a necessity to provide sufficient heated air to dry the product consistently within shortest period. Under solar energy, drying process takes minimum of 2 days but with low quality of products

and exposure to air contamination. Whereas, under the heatroom system powered with a combination of sunlight, gas, electricity utilities, the pace of drying times is a matter of hours with high-quality product output.

Shukrah Food Mills designs a drying technology with radiating heated air in dehydrators and warehouses that ensures both high volume and quality products within short drying times. The objective of this system is to achieve continuous production under any weather condition, which is difficult with the sunlight energy alone. This warehouse system saves a lot of alternative artificial energy resources when the sunlight is consistently available. Thus, drying production system is expected to be consistent, cheap to operate and available within any rural community. It is expected to utilize all optional energy sources including sun, gas, fuel, electricity, or firewood.

The power system has its base or unit outside the walls of the warehouse while the heated air flows and circulates into the interior of the warehouse. The store is constructed with stackers of long rows. Each stack measures 4ft wide surface and can be if available length of the warehouse. The tray lines are fixed to frames. Subject to financial capability, the trays lines and frames can be made of steel, wooden products, or a combination of both. The walls of the storage must be airtight so as not to allow heated air to escape thereby slowing down drying times. Again, with enough funds, the interior of the storage can be padded with polythene or tarpaulin or PVC materials.

The energy source consists of a thick steel plate taken as a burner. This burner is placed on the heat energy like it will cooking pot. The heat generated is circulated by air radiating within the warehouse and thus penetrates the various steps of trays to dry the products spread on the shelves. A fan system can be installed to aid the circulation of the air. Air vents are also added at the extreme upper ends of each side of the warehouse to aid the cooling system when products are fully dried to prevent over drying.

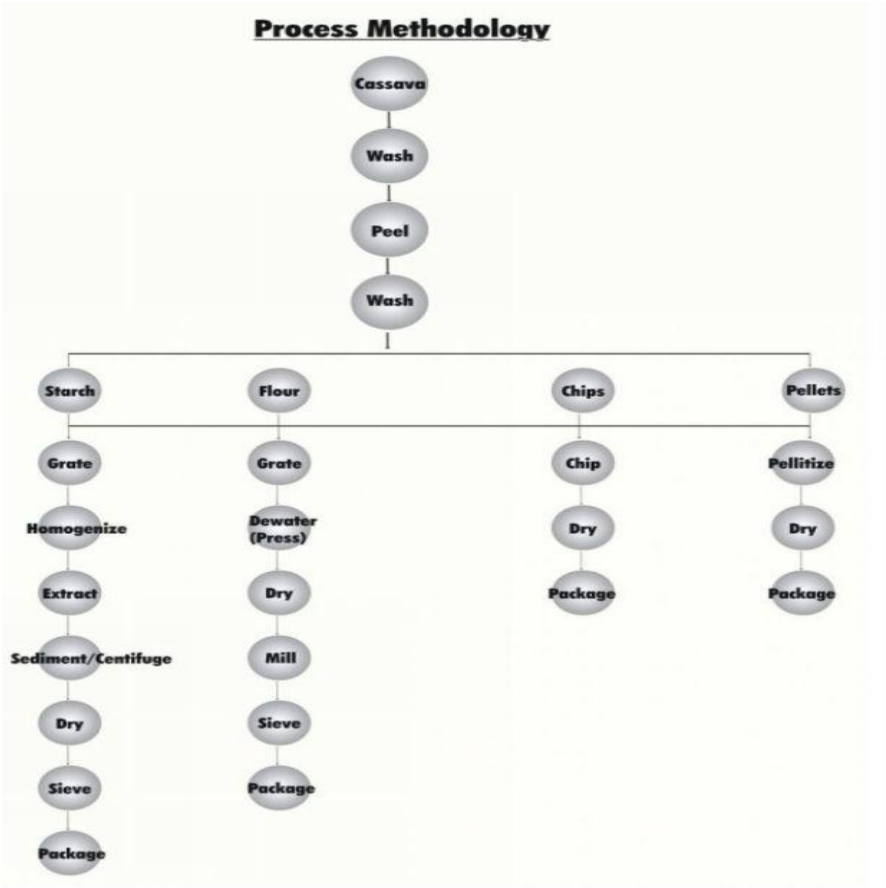


Heat room Storage facility

3.3 DRIED CASSAVA PRODUCTION

Drying is the process by which much of the water contained in cassava roots is eliminated, thus obtaining a dried product that can be stored over a long period.

Drying can be either natural or artificial and is carried out either on farms or in industries. Various scales of operation exist, differing as much in their costs and levels of technological sophistication as in the end uses of their products.

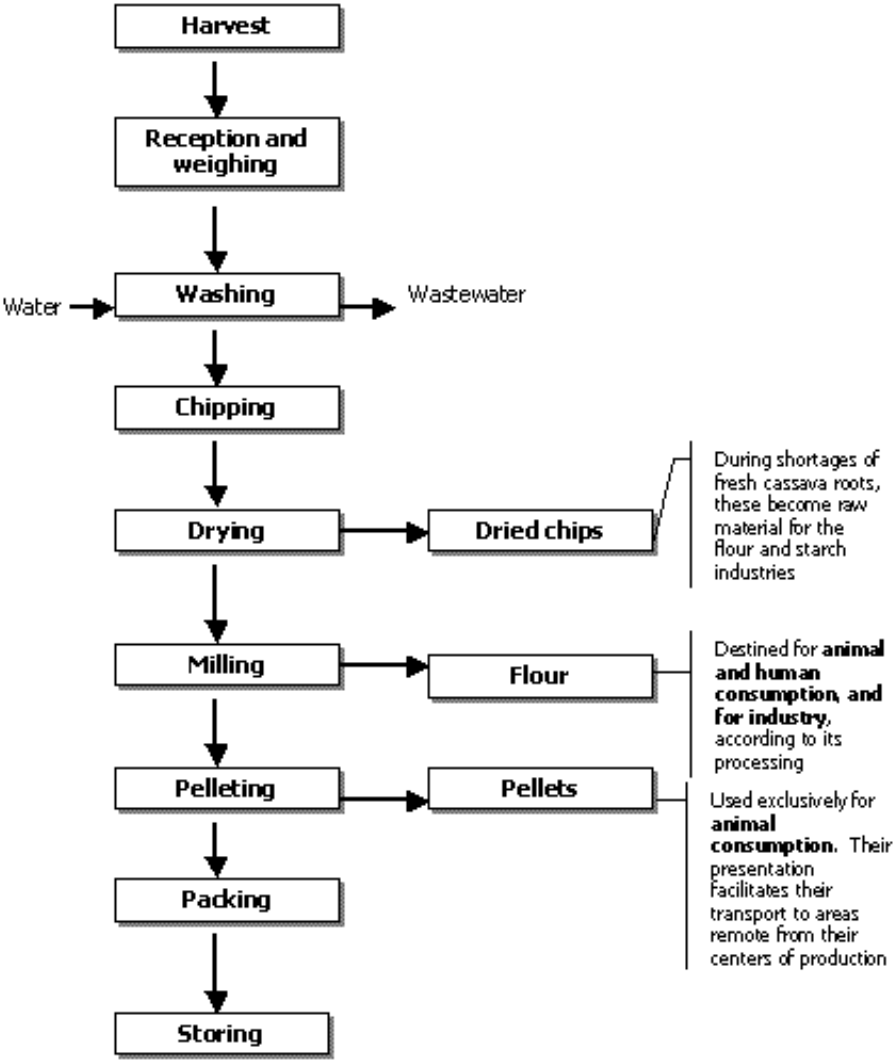


Natural drying:

There are two common levels of natural drying on concrete floors namely small and large scale. According to findings around the world, small-scale drying was first developed in Thailand, with CIAT adopting it during the 1960s for use in Colombia. This technology is also used in African countries, Brazil, Ecuador, Indonesia, Malaysia, Thailand, and Vietnam.

This method takes advantage of solar energy and open air. The cassava is chipped, then dried on concrete floors. Small-scale

processing plants characteristically have drying areas between 500 and 2000 m².



3.4 SMALL SCALE DRYING PRODUCTION

3.5

	**Reception and weighing** Product is weighed after peeling to determine its yield.
	**Washing** this operation is done after peeling of the products and before parboiling in the case of yam and plantain or banana.
	**Chipping** Small and uniform chips are made for rapid and homogeneous drying.
	**Spreading** The chips are spread in layers of 10 to 12 kg of chips per m² (=12 x 1800) or 2160kg per plot of land or 43 bags of 50kg bags. Or simply 2 tons of products.
	**Turning over** The chips are turned over every 2 hours (i.e., 6 to 8 times a day) to ensure uniform drying.



Collecting and packing

Once the chips have reached a moisture content of 10% to 12%, they are placed in 40 or 50-kg polyethylene sacks.



Storing

The dried chips (at 10% to 12% moisture content) can be conserved for 6 to 12 months without their quality being endangered. Optimal conditions for storage are a storeroom with good ventilation, low moisture, and adequate hygiene.

Inspection

About 38 species of insects, mostly of the order Coleoptera, have been found in dried cassava chips. However, the important ones are those that can reproduce among the chips, such as *Araecerus fasciculatus* (coffee bean weevil), *Dinoderus minutus* (bamboo borer), *Lasioderma serricorne* (tobacco beetle), *Latheticus oryzae* (long-headed flour beetle), *Stegobium paniceum* (drugstore beetle), and *Tribolium castaneum* (red flour beetle).

3.5 LARGE-SCALE DRYING PRODUCTION

This technology is used in countries such as Indonesia, Malaysia, and Thailand. This last country is the world's principal producer and exporter of cassava chips, exporting mostly to the European Union. World production surpasses 15 million tons per year, which nevertheless reflects a potential that is currently underused.

A processing plant characteristically has a drying area between 20,000 and 70,000 m².

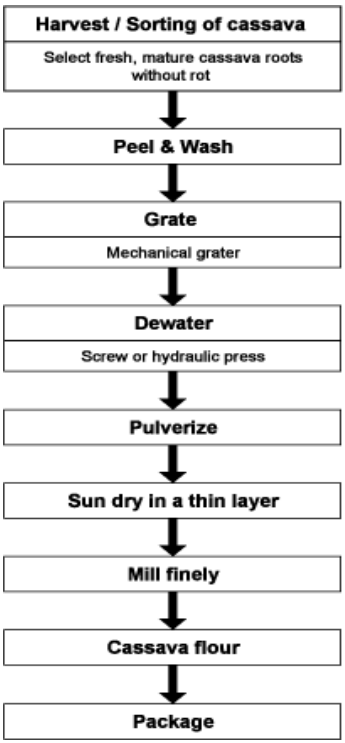
	**Reception and weighing** Cassava is weighed before and after drying to determine yield.
	**Chipping** The chipping machines can process as much as 30 tons per hour.
	**Spreading** Tractors and bulldozers are used to spread the chips over the concrete floors.
	**Turning over** The chips are turned over with special equipment attached to a vehicle. If rain is likely, the chips are collected

	or covered.
	**Collecting and packing** The pieces are collected, using tractors or bulldozers.
	**Storing Transport** The dried chips are transported to pellet plants, which can store 10,000 to 20,000 tons of chips in bulk.

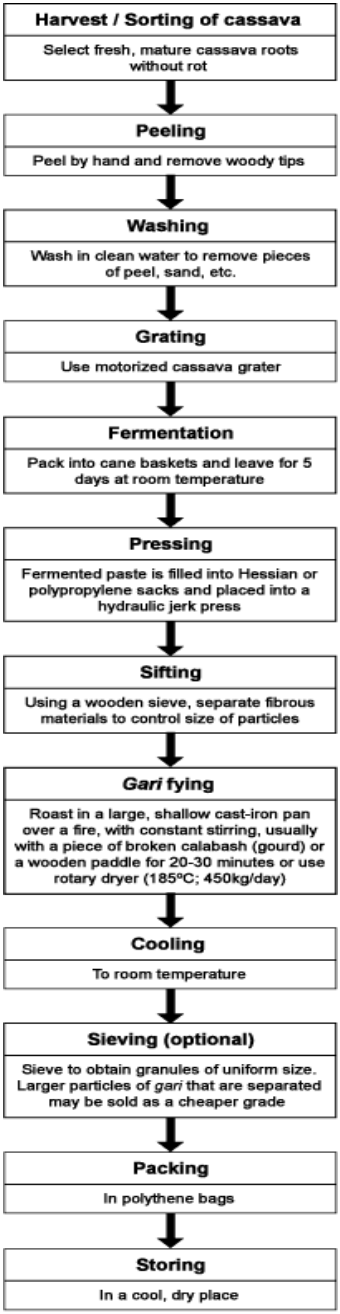
3.6 CASSAVA FLOUR PRODUCTION

The units of operations involved in cassava flour process technology are:

1. *Sorting:* The freshly harvested tubers are sorted mainly by visual assessment. Defective tubers and extraneous matters are separated and discarded.
2. *Weighing:* The sorted tubers are weighed with industrial weighing scale. The total weight of the input (raw material) can be recorded.



3. *Peeling:* The separation of inedible peels from the edible portion of the tubers is accomplished by manual peeling. The peels can be utilized for poultry feed formulation or serve as source of fuel for industrial plants.
4. *Washing:* The peeled roots are thoroughly washed with hygienically clean water to remove both physical and biological contaminants, which can adversely affect the quality of the finished product. In addition, free starch is released by the cutting action of the peeling knife or brushing during harvesting. Washing can be accomplished manually or mechanically (using a drum washer).
5. *Grating:* After weighing (to calculate the peeling loss) the peeled tubers are subjected to the size reduction unit operation called grating. Ideally, a hammer mill is used to produce a smooth mash, which would result in a superior quality product with higher yield.



6. *Fermentation:* The mash is aerobically fermented in a non-corrosive container for a very short period, (2hrs). From zero hours to the end of the period, the mash is intermittently and vigorously stirred manually or mechanically. This ensures intimate contact of the enzyme (endogenous) with hydrolysis of the cyanogenic complex and maximum liberation of the severed hydrogen cyanide. The process is termed detoxification.
7. *Dewatering:* Immediately after the fermentation period, the mash detoxified is dewatered from about 75% moisture content to less than 50%. Several systems are available such as of those in use at FIIRO manual screw press, hydraulic press and basket centrifuge. The resultant product is termed "CAKE"
8. *Granulating*: The semi-solid wet "cake" is mechanically size – reduced using a hammer mill granulator. Granulation produces fine granules with greater surface-area than cake.
9. *Dehydration:* Dehydration (or drying) of the fine granules is accomplished using a mechanical (rotary hot air) dryer rather than sun drying. Owing to increase in the surface area of the granules the drying is much faster. Mechanical drying ensures improved and consistent product quality. The flash dryer is equally good equipment for drying cassava flour. The moisture content of the product is about 8-10%.
10. *Milling:* The dried product is milled with a hammer mill to produce a fine powder.
11. *Sieving:* The powder is sieved to separate the fibre from the required fine flour (a creamy-white product).

12. *Packaging:* The flour produced is packaged directly or blended with flour before packaging e.g. Wheat flour to form composite flour.

3.7 DRYING SYSTEM

Under optimal conditions (dry sunny weather for sun drying), the IITA technique enables small-scale primary processors to produce high-quality unfermented cassava flour that meets the specifications of industrial users within one day. Drying has been identified as the major tool for expanding processing of cassava into high-quality cassava flour. Various options have been considered so far in the Cassava project at IITA.

Natural drying

Local processors expose cassava mash on a polythene sheet directly to the sun. This is referred to as "sun drying". The project observed that drying at rural or domestic levels cannot be done artificially because of the high capital investment in equipment and energy required and hence, natural sun drying is done.

Sun drying is beset by several inherent drawbacks, such as susceptibility to damage due to inclement weather, slow drying rates, and contamination. Because of these limitations and the high cost and low utilization of more efficient traditional dryers, the adoption of a modified sun drying process, solar drying, has been considered for drying HQCF in rural areas.

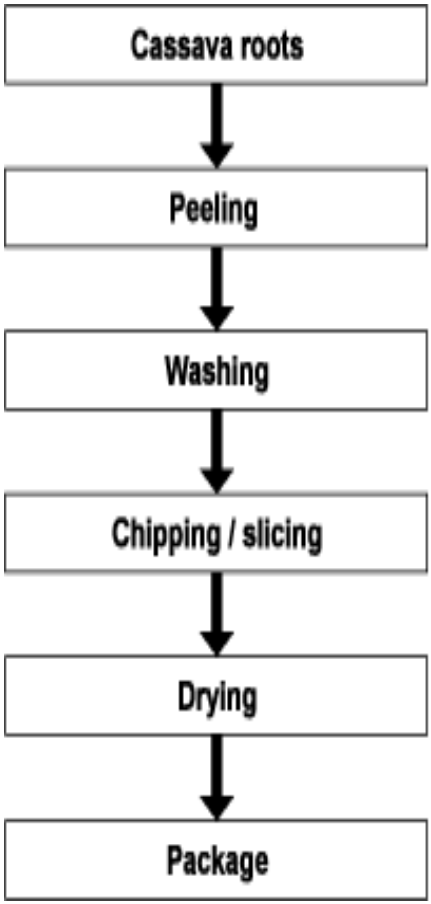
Artificial drying

If a controllable source of energy is used for drying operations, the process is referred to as artificial or

mechanical drying. There is a further classification in which the air used for drying is heated, either by solar means or controlled means such as electricity, renewable fuels, or fossil fuels. These methods are referred to as hybrid drying.

Rotary dryer

IITA is building up a coalition of local fabricators to mass produce an already tested locally fabricated rotary dryer for use by the rural processors of high-quality cassava flour. The dryer could be fired by charcoal or gas and rotated by a diesel engine. The dryer consists of an insulated drying chamber (drum-like). The dryer is cost effective and user friendly.



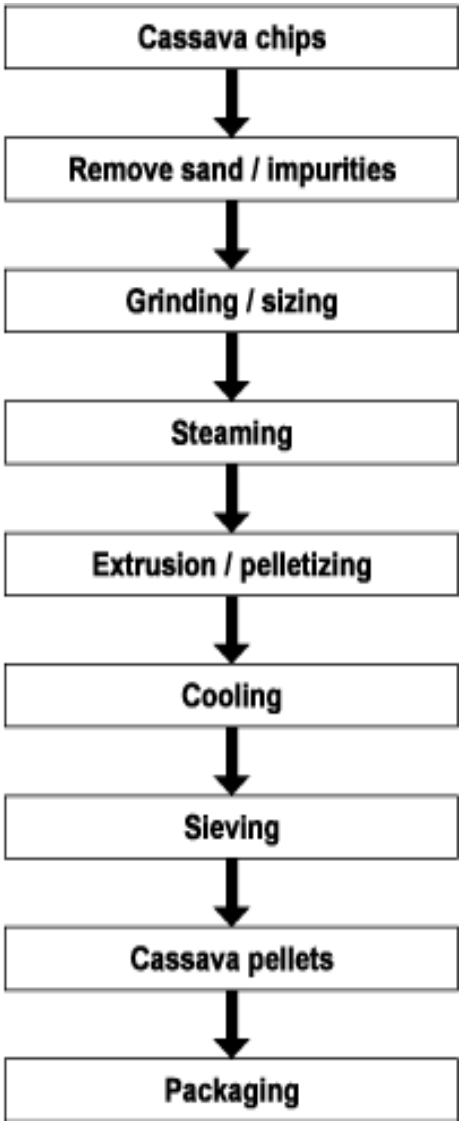
Flash dryer

IITA modified the design and fabrication of a locally fabricated flash dryer with Peak Products, Abeokuta, Nigeria. The flash dryer (3t/day capacity) is well insulated with the product contact surface made of stainless steel and a semi-automated feeder. It could be operated with combined kerosene/spent oil. The dryer is user friendly and will encourage greater cassava flour production.

3.8 GARRI PRODUCTION

Gari is creamy-white, granular flour with a slightly fermented flavor and a slightly sour taste made from fermented, gelatinized fresh cassava tubers. Gari is widely known in Nigeria and other West African countries.

It is commonly consumed either by being soaked in cold water with sugar, coconut, roasted ground nuts, dry fish, or boiled cowpea as complements or as a paste made with hot water and eaten with vegetable sauce. When properly stored, it has a shelf-life of six months or more.



Raw material

Fresh cassava should be free from microbial or insect damage and without serious bruising or cuts. It should be processed within two days of harvest to prevent deterioration and loss of quality in *gari*.

Hygiene

Fresh cassava is a moist, low-acid food that is susceptible

to bacterial and fungal growth. Hygienic practices, especially in the early stages of processing, should ensure minimal contamination. All waste materials from the process should be removed from the site as they are produced to avoid the risk of cross-contamination.

Process control

Washing should be carried out thoroughly to avoid contamination of the final product with peel, sand, and so on. Fermentation must be properly controlled, as a period too short will result in incomplete detoxification and a blend product. Too long a period will give the product a strong sour taste. Both over- and under fermentation also badly affect the texture of the final *gari*.

If too much liquid is pressed from the grated cassava, the gelatinization of starch during subsequent roasting is affected and the product is whiter.

If sufficient liquid is not removed, however, the formation of granules during roasting is affected and the dough is more likely to form into lumps. The ideal moisture content is 47-50%, and experienced gari producers assess this visually. Sieving is important to obtain a high-quality product, free of fibrous contaminants and with similar-sized granules. The granules must be roasted to about 80ºC/175 ºF to achieve partial gelatinization of the starch. If lower temperatures are used, the product simply dries and produces a dry white powder. Too high a temperature will cause charring of the product and make it stick to the roasting pan.

Packaging and storage

The product is hygroscopic (it absorbs moisture from the air) and should be packed in airtight and moisture-proof bags, especially in areas of high humidity, to prevent mold growth.

Processing Equipment

i. Weighing machine
ii. Washing machine
iii. Presser
iv. Grater
v. Granulator
vi. Sieves
vii. Fryers/Rotary Dryer
viii. Hammer milling machine
ix. Sealer

Grating of cassava roots after 3 days of submerged fermentation using a mechanical grater.

A second stage of solid-state fermentation (this can be described as “double fermentation”). Sieving with very little water.

3.9 CASSAVA CHIPS PRODUCTION

This is the most common form in which dried cassava roots are marketed, and most exporting countries produce them. The chips are dried irregular slices of roots, which vary in size but should not exceed 5 cm in length so that they can be stored in silos.

They are produced extensively in Thailand, Malaysia, Indonesia, and some parts of Africa. The present method of processing chips is very simple, consisting of peeling, washing,

chipping the cassava roots, and then sun drying the slices or chips.

The recovery rate of chips from roots is about 20-40% depending on the initial dry-matter content of the cassava roots and the final moisture of the chips.
General specifications for cassava chips, flour, starch, and pellets are presented in while specifications for chips intended for export to the European Community for use in livestock feeds is in.

Chips should be white or near white in color, free from extraneous matter, molds, insect infestation and damage, and possess no peculiar odors. In addition, shipments of chips must not contain significant amounts of dust, as this is considered unacceptable by European importers.

Broken roots
Like chips in appearance, but generally thicker and longer, they are often 12-15 cm long and can jam the mechanism of handling equipment. They are produced mainly in Africa where local processors prefer to produce longer roots because of the domestic demand mainly for products suitable for human consumption, as cassava is part of the staple diet. Once processed into chips, the product becomes inedible, and the producer wants to conserve the local market.

3.10 CASSAVA PELLETS PRODUCTION

Pellets are obtained from dried and broken roots by grinding and hardening into a cylindrical shape. The cylinders are about 2-3 cm long and about 0.4-0.8 cm in diameter and are uniform in appearance and texture.

Pellets are produced by feeding dried cassava chips into the Pelleting machine, followed by screening and bagging for export. Powdered chips, which fall during Pelleting, are re-pressed into pellets and the process repeated. There is usually about 2-3% loss of weight during the process. Pellets have the following advantages over chips: quality is more uniform; they occupy 25-30% less space than chips, thus reducing the cost of transport and storage; handling charges for loading and unloading are also cheaper; they usually reach their destination sound and undamaged, while a great part of a cargo of sliced chips is damaged in long-distance shipment because of sweating and heating.

Shukrah Food Mills at Oyo Town has been able to design and fabricate a series of small-scale industrial dehydrating machines that can produce different types of dried crops such as cassava, plantain, yam, etc and pellets used for animal feeds.

3.11 FIIRO CASSAVA PELLETS PRODUCTION METHODS

The production process comprises simple operations, which have been mechanized. The equipment used can be fabricated in the Institute. The process flow chart is as attached in figures A and B.

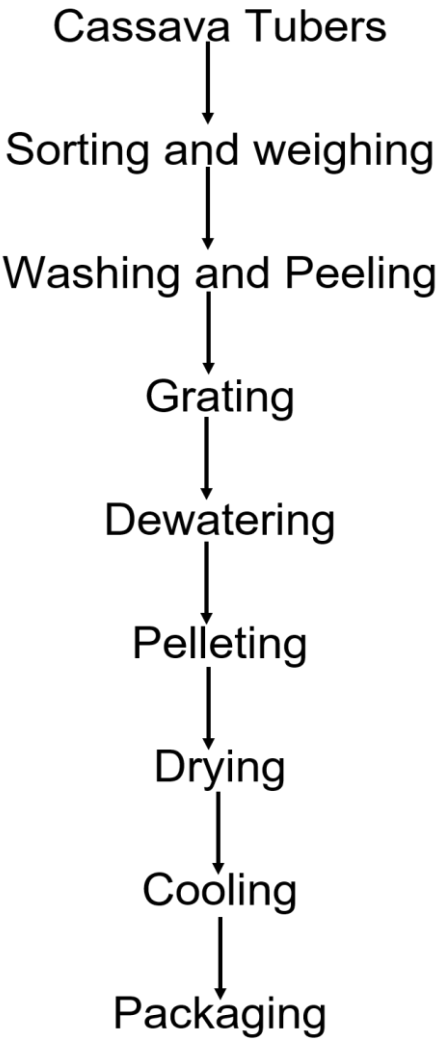


Figure A:
Production of cassava pellets from raw cassava.

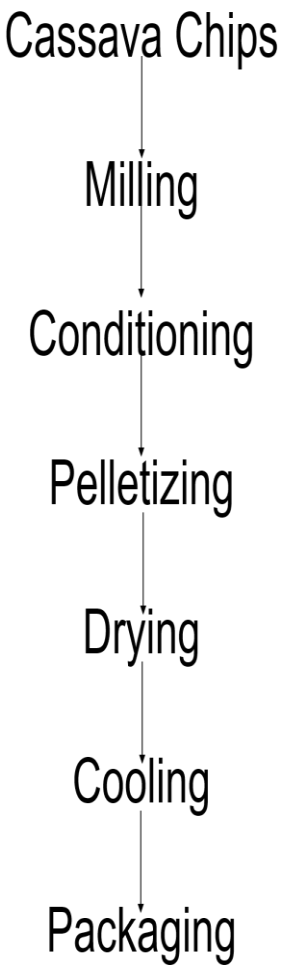
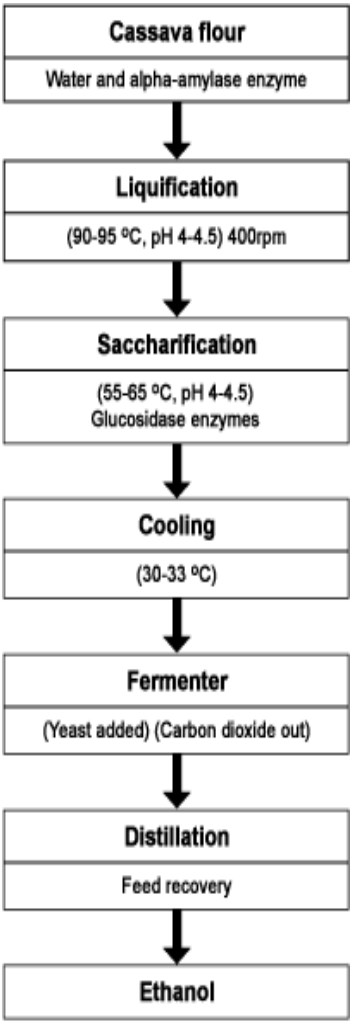


Fig. B:
Production of Cassava pellets from chips

3.12 ESSENTIAL OPERATIONS ARE AS FOLLOWS:

1. *Sorting and weighing:* The raw cassava is sorted out to remove unwholesome roots and foreign materials before weighing.
2. *Washing and Peeling:* The cassava roots are washed in potable water and manually peeled with knives.
3. *Grating:* The peeled roots are milled into mash using grater.
4. *Dewatering:* The wet mash was dewatered to a suitable moisture content using a hydraulic press.
5. *Pelletizing:* The dewatered mash is then fed into a pelletizing machine to form pellets of predetermined shape and size.
6. *Cooling:* The pelletized product is allowed to cool to ambient temperature prior to packaging.
7. *Packaging:* The pelletized product is packaged in light density polyethylene bag or double lined polyethylene bags.



3.13 ETHANOL PRODUCTION

Ethanol from Cassava

Ethanol is generally produced by the fermentation of sugar, cellulose, or converted starch and has a long history. In Nigeria, local production of ethanol from maize, guinea corn, millet, other starchy substrates, and cellulose is as old as the country itself. Apart from food and pharmaceutical uses, ethanol finds itself alternative use for biofuel in most of the developed world for the following reasons:

Cassava feeds for livestock other than poultry
Several trials using cassava on cattle, sheep, goat, and pigs are reported in Hahn et al. (1992) and recently by Tewe (2003).

- It is not poisonous.
- It does not cause air pollution or any environmental hazard.
- It does not contribute to the greenhouse effect problem (CO_2 addition to the atmosphere, causing global warming).
- It has a higher-octane rating than petrol as a fuel. That is, ethanol is an octane booster and anti-knocking agent.
- It is an excellent raw material for synthetic chemicals.
- Ethanol provides jobs and economic development in rural areas.
- Ethanol reduces country's dependence on petroleum, and it is a source of non-oil revenue for any producing country.
- Ethanol can reduce the adverse foreign trade balance.

3.14 ETHANOL PLANT

Basic plant

The first type of plant will produce strong alcohol from cassava, but this will have an odor as the distilling process is very crude. The plant would bring in fresh cassava, wash and peel, grate, cook in a jet cooker, ferment, distil, and bottle. In addition, a steam boiler, generating set, effluent treatment plant and electrical system are required. The actual amount of cassava needed is dependant upon the starch content, but as a guide, cassava at 30% starch content will produce approximately 280 liters of alcohol/ton. Cassava, which is only 20% starch, will produce only 180 liters of alcohol/ton. The plant will produce approximately 3000 liters/day of alcohol at 96%, which is equivalent to 7500 liters/day at 40%. The plant will also produce around 2-3m3/hr of effluent. This must be disposed of properly and is normally used as animal feed. This plant operates on a batch basis and can process approximately 4-6 batches/24 hours, producing 500 liters of alcohol/batch. The plant will need good water supply and continuous electrical supply (around 50Kva). The steam requirements are around 1500 kg/hour.

The cost will be in the region of £290,000, which includes all the equipment, shipping, supervision of installation, commissioning, and operator training.

Complex plant

The next type of plant will produce a better quality of alcohol from cassava, and uses a multi-column still so that the complex distilling techniques employed ensure a high-quality product, free from all types of odors, etc. The plant would

bring in fresh cassava, wash and peel, grate, cook in a jet cooker, ferment, distil, and bottle. In addition, a steam boiler, generating set, effluent treatment plant, and electrical system are required. The actual amount of cassava needed depends upon the starch content, but as a guide cassava at 30% starch content will produce approximately 280 liters of alcohol/ton. Cassava, which is only 20% starch, will produce only 180 liters of alcohol/ton. The plant will produce approximately 4000 liters/day of alcohol at 96%, which is equivalent to 10,000 liters/day at 40%. The plant will also produce around 2-3m3/hr of effluent. This must be disposed of properly and is normally used as animal feed. This plant operates on a batch basis and can process approximately 4-6 batches/24 hours, producing 500 liters of alcohol/batch. The plant will need a good water supply and a continuous electrical supply (around 50Kva). The steam requirements are around 1500 kg/hour. The cost will be in the region of £620,000, which includes all the equipment, shipping, and supervision of installation, commissioning, and operator training.

3.15 LAFUN PRODUCTION

Lafun is a fibrous powdery form of cassava like *fufu* in Nigeria. The method of production of *lafun* is different from that of *fufu*. In the traditional preparation, fresh cassava roots are cut into chunks and steeped for 3-4 days or until the roots become soft.

The fermented roots are broken up into small pieces, and sun dried on mats, flat rocks, cement floors, or the roofs of houses.

The dried pieces are milled into flour. Alternatively, chips are made directly from fresh roots, cut into chunks, and sun dried. Drying takes 2-4 days, depending on the weather.

Unlike *fufu*, the fibers in the retted root for *lafun* are dried along with the mash and later sieved out. Thus, lafun is coarser than *fufu* in texture. The flour is made into dough with boiling water before consumption. When properly stored, it has a shelf life of six months or more.

3.16 YAM FLOUR (ELUBO) PRODUCTION

Elubo is the Yoruba term for peels of yam, which are dried and ground into brown flour. It is basically flour made from yam, yam or cocoyam, and eaten much like mashed potatoes. Either of these two varieties is made by:

1. Selecting and washing the raw yam tubers
2. peeling the yam,
3. Boiling clean water at about 100°C.
4. Dipping and leaving to cool out with the hot water for about 8 hours
5. Spread the product to air dry and further dry until it becomes crisp
6. Grinding the crisp in a milling machine.

Sometimes, the skin or peelings of yam is used. This would even be richer in fibre content with less starch and become an essential source of protein for animal feeds.

3.17 POUNDED YAM FLOUR

Pounded Yam Flour (PYF) brings succor to pounded yam lovers as the drudgery of pounding is eliminated.

- *Yam selection and weighing*: Wholesome suitable varieties are selected for PYF production. The wholesome tubers are then weighed.
- *Washing:* The selected tubers are washed properly to remove adhering sand.
- *Peeling and slicing:* The washed tubers are carefully peeled manually with stainless steel knives. The peeled tubers are then mechanically sliced or chipped with machine operations.
- *Parboiling:* The yam slices are blanched in boiling water (parboiled) for about 20 minutes, depending on the effectiveness of the heat/fire source.
- *Drying:* The blanched product is then dried in a dryer for 6 to 8 hours at a temperature of 122° to 158° F (50° to 70°C) until the pulp is about 10% moisture free. The dried yam chips are stored in airtight containers.
- *Milling*: The dried products are pulverized (milled) directly into flour.
- *Packaging:* The dried products are packed into airtight, moisture- proof packaging materials

3.18 FLASHCARDS: TECHNOLOGY DEVELOPMENT

1. **Q:** What is Drying?
   A: The process of reducing moisture content to extend shelf life and preserve products.
2. **Q:** What is Natural Drying?
   A: Product drying using solar energy and natural air circulation.
3. **Q:** What is Mechanical Drying?
   A: Product drying using machines, controlled heat, and forced airflow.
4. **Q:** What is the purpose of drying cassava chips?
   A: To reduce moisture and prepare chips for storage, milling, or industrial processing.
5. **Q:** What are the major stages in cassava flour production?
   A:
   - Harvesting
   - Washing
   - Peeling
   - Chipping
   - Drying
   - Milling
   - Packaging
6. What are the benefits of mechanical drying?
   A:
   - Faster drying
   - Better quality
   - Reduced contamination

- Higher productivity

Multiple Choice Questions

DSS stands for:
a) Distributed Storage Service
b) Direct Supply System
c) Drying Security Service
d) Distribution Sales Strategy
Answer: a

1. The primary objective of DSS is:
a) Tax collection
b) Preserve fresh products
c) Product export
d) Product branding
Answer: b

1. Cassava chips are generally dried to:
a) 50% moisture
b) 30% moisture
c) 10-12% moisture
d) 25% moisture
Answer: c

1. Gari production requires:
a) Fermentation
b) Roasting
c) Sieving
d) All of the above
Answer: d

1. Mechanical drying improves:
2. a) Product quality

3. b) Consistency
4. c) Production speed
5. d) All of the above

Answer: d

Short Answer Questions

1. Describe the Drying Production System.
2. Differentiate between natural and artificial drying.
3. Outline cassava flour production processes.
4. Explain gari production control requirements.
5. Describe cassava pellet production.

Practical Questions

Question 1, Draw a process flowchart for Cassava Flour Production.

Question 2, Develop an SOP for:

- Washing
- Peeling
- Drying
- Milling
- Packaging

Question 3, Compare:

Factor	Sun Drying	Mechanical Drying
Cost		
Speed		
Quality		
Output		

CASE STUDY

Design a drying warehouse capable of processing 5 tons of cassava chips daily.

Calculate:

- Drying area required
- Heat requirements
- Labour needs
- Storage capacity
- Estimated operating costs

CHAPTER FOUR

SALES AND DISTRIBUTION MANAGEMENT

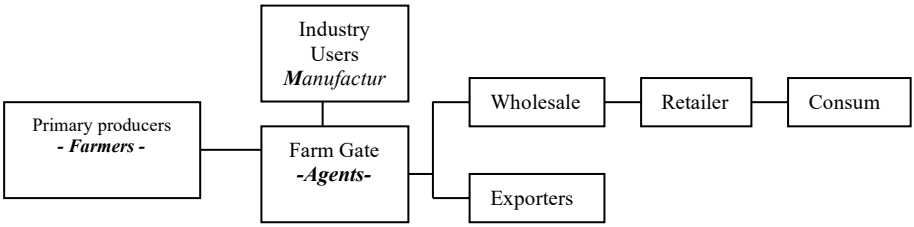
4.1 Introduction

By a critical study, dried food products are inputs for consumption or industrial raw materials. The consumer products include gari, lafun and fufu for direct cooking and eating. The industrial raw materials included flour, pellets, chips/chunks, and industrial starch. Households and families are more on the consumer producer group operating at micro scales. The industrial producers are presently scanty and operate at micro, small and medium scales. There is a great need for investment intervention to fill the gap between the consumer producers and the industrial users. All products lines have little or no differences in the sales, marketing and distribution chains. Dareslauf operates at 60% in the first year to increase to 80% in the following two years.

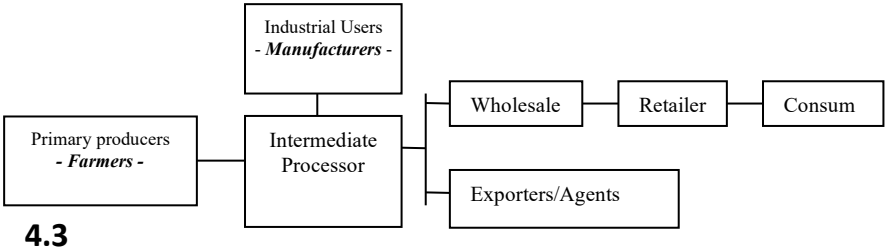
4.2 The market structures

The marketing chain identifies the different stages between production and final consumption. The market consists of eight functions: production, drying, stocking, wholesaling, milling, packaging, exporting and retailing. The degree of the industrial sub sector can be described as a perfect competitive. Buyers and producers are all well-informed about the industry. The size and distribution of buyer and seller markets are key determinant factors for the strength of any participant in the industry. There is presently no control over the producers and marketers' independence in Nigeria.

The present supply and marketing chains



The planned supply and marketing chains



4.3

4.4Marketing-production strategies

The nature of the various raw products has a lot of implications for the facilities required to market at different stages. These include haulage cost, storage space, and risk. The food chain includes gari, lafun, fufu, starch, pellets, chips, chunks and other value-added industrial products. Each of these has different traditional methods of production. The methods of production by Dareslauf are mechanized to achieve high quality of food health, safety, environment, and security and above all for different nutritious dietary provisions. The marketing strategies are to identify ways of adding value to products to exploit the intermediate segment of the marketing chain.

The production system is to identify the operations as a secondary processor serving the retailers, wholesalers, and industrial users as well as exporters alike. The marketing

system is to move the products to the urban centres for the identified market segment above exploiting the full opportunities for primary, intermediate and users. The system will ensure at least 50 – 70-% of the benefits in the marketing chain is retained.

The critical facilities of the marketing chains that would ensure better prices are identified are transportation, advertisement, distribution networks, communication, grading, branding/packaging, storage, etc. the target destinations are the urban populated locations and allied industries

4.5 Capacity metrics

The present design is to achieve 1.5ton/week total products output. The startup capacity is 60% in the first year of operation and to increase by 10% in each subsequent year till the third year of operation. All associated key elements of operation (revenue, personnel, and procurement) are to be managed at this level of operation given the micro and macroeconomic prevalence.

The operation is to integrate labour, materials, machines, and logistics of purchasing, storage, and delivery of products. All operational activities involved to turning out final products are to be identified in line with the key operational units in the factory setup. The potential input products in the production line are identified to include yam, cassava, plantain, banana, potato products. All procurements and logistical operations are to be organized in line with the production requirements determined by production capacity

and orders for the various products. The key operational units of the factory are:

purchasing unit,
Production unit,
Sales/marketing unit, and
Administration unit.

The tables in section six below contain the guide for planning the operations



4.5 FLASHCARDS - SALES AND DISTRIBUTION

1. **Q:** What is Value Addition?
   A: Improving the form, quality, packaging, or usefulness of a product to increase its value.
2. **Q:** What is a Marketing Chain?
   A: Activities involved in moving products from producers to final consumers.
3. **Q:** List the main marketing channels.

   A:
   Farmers
   Processors
   Wholesalers
   Retailers
   Exporters
   Final Consumers
4. **Q:** Why is packaging important?
   A: Protection, branding, transportation, shelf appeal, and product information.
5. **Q:** What are target markets?
   A:
   Households
   Hotels
   Bakeries
   Food industries
   Livestock feed manufacturers
   Export buyers
6. **Q:** What is Distribution Management?

A: Planning and controlling the movement of products from manufacturers to customers.

Multiple Choice Questions

1. The marketing chain contains
   a) Three functions
   b) Five functions
   c) Eight functions
   d) Ten functions
   Answer: c

2. Which is NOT part of the marketing chain?
   a) Production
   b) Exporting
   c) Mining
   d) Packaging
   Answer: c

3. Major target markets include:
   a) Households
   b) Bakeries
   c) Industrial users
   d) All of the above
   Answer: d

4. Capacity utilization in year one is planned at:
   a) 40%
   b) 50%
   c) 60%
   d) 90%
   Answer: c

5. Value addition improves

a) Profitability
b) Marketability
c) Product differentiation
d) All of the above

Answer: d

Short Answer Questions

1. Explain the Dry Harvest marketing chain.
2. Discuss the role of packaging in agro processing.
3. Describe key marketing facilities needed.
4. Explain industrial-user marketing strategy.
5. Define product distribution network.

Practice Assignment

Prepare a marketing plan for:

- Cassava Flour
- Yam Flour
- Plantain Flour

Include:

- Target market
- Pricing strategy
- Distribution channels
- Promotional activities
- Sales forecast

Essay Questions

1. Discuss the importance of branding and packaging in agro processing.
2. Evaluate marketing strategies for reaching urban and export markets.

CHAPTER FIVE

HYPOTETIC CASH FLOW PROJECTION

5.1 INTRODUCTION

A standard factory operation deserves production planning and cash flow appraisal. This is primarily a requirement to get the financial status of operating the facility during a specified period. And indeed, the bottom line is to be guided on the profit and loss levels to secure financing. Presented below are typical analysis and cash flow projection.

Labour Estimate

Positions	Purch	Prod	Sales	Admin	Total
Officers		1			1
Assistants	1	1	1	1	4
Operators		2			2
Casual Workers		2			
Total	1	4	1	1	7

5.2 Salaries and Wages

Salaries	N6,000	N10,000x1, N6,000x 1 N8,000x2, 32,000	N6,000	N6,000	N50,000
Casual Wages (2 shifts) 250x2x6x4x2		N24,000			N24,000
Monthly Total	6,000	56,000	6,000	6,000	74,000
Yearly Total	72,000	672,000	72,000	72,000	888,000

Production Overseas

Product	Diesel, Oil & Gas	Transport/ Admin Exp	Packaging	Total
Weekly	36,000	55,000	30,000	121,000
Monthly	144,000	220,000	120,000	484,000
Yearly	1,728,000	2,640,000.	1,440,000	5,808,000

5.2 PRODUCTION ANALYSIS

i. ***100% Capacity Estimate*:**

Concrete Platform1 = 270kg Dried product

Concrete Platform2 = 270kg Dried Product

Heat room = 648kg Dried Product

Total = **1,188kg** Dried Product

ii. Production Estimate

Production Runs per week	=	2 times
Weekly Production	=	2,376 kg
Dried Products		
Production per month	=	9,504 kg (100%)
60% Planned capacity	=	5,702kg
Input/ Output Ratio	=	2:1
Input Product@60%	=	11,405kg
Raw Products		

iii. Sales and Purchases Estimate

Cost per ton of Raw products = N30,000

Sales per ton of Dried Product = N150,000

Cost of Raw Materials per month = N342,150

Sales of Dried Product per month = N855,300

iv. Debtors and Creditors Estimate

Debt servicing = 75% of monthly accruals

Creditor balance = 25% of Input Cost and Overheads

Debt Recovery = 50% of Sales in the month in 45 days

Debtors Balance = 50% of Sales in the month

v. Stocks:

Raw Product = 2,851kg@N30,000/ton

= 85,530

Finished Products = 1,426@150,000/ton

= 213,900

= 299,430

5.3 SIX MONTHS PROFIT AND LOSS PROJECTION

Months	1	2	3	4	5	6	Total
Sales/Turnover	**428**	**855**	**855**	**855**	**855**	**855**	**4,704**
Direct Cost:							
Raw Materials	342	342	342	342	342	342	2,053
ılaries and Wages	44	44	44	44	44	44	266
Production Overheads	290	290	290	290	290	290	1,742
	677	**677**	**677**	**677**	**677**	**677**	**4,062**
G. Profit/(Loss)	-249	178	178	178	178	178	642
Ops Cost/ton	119.0	119.0	119.0	119.0	119.0	119.0	119.0
G.Profit/Loss per ton	-432	32	32	32	32	32	32
Sales per ton	150	150	150	150	150	150	150
G. Profit/(loss)	(37%)	26%	26%	26%	26%	26%	26%
Profit/Loss Sales	(58%)	21%	21%	21%	21%	21%	21%

Account Receivables

	₦'000	₦'000	₦'000	₦'000	₦'000	₦'000	₦'000
Sales	428	855	855	855	855	855	4,704
Receipt on Acct	-214	-642	-642	-642	-642	-642	-3,421
Difference	214	214	214	214	214	214	1,283
Balance B/F	0	214	428	641	855	1,069	0
Balance C/F	**214**	**428**	**641**	**855**	**1,069**	**1,283**	**1,283**

Account Payables

	₦'000	₦'000	*₦'000*	₦'000	₦'000	₦'000	₦'000
luct Supplies	342	342	342	342	342	342	2,054
erheads/Exp	290	290	290	290	290	290	1,742
Total	633	633	633	633	633	633	3,795
nent on Acct	-474	-474	*-560*	*-560*	*-560*	*-560*	-3,274
Difference	158	73	73	73	73	73	521
Balance B/F	0	158	*231*	303	376	449	0
Balance C/F	**158**	**231**	***303***	**376**	**449**	**521**	**521**

Account Payables

	₦'000	₦'000	*₦'000*	₦'000	₦'000	₦'000	₦'000
oduct Supplies	342	342	342	342	342	342	2,054
verheads/Exp	290	290	290	290	290	290	1,742
Total	633	633	633	633	633	633	3,795
yment on Acct	-474	-474	*-560*	*-560*	*-560*	*-560*	-3,274
Difference	158	73	73	73	73	73	521
Balance B/F	0	158	*231*	303	376	449	0
Balance C/F	**158**	**231**	***303***	**376**	**449**	**521**	**521**

Receipt and Payment

	₦'000	₦'000	*₦'000*	₦'000	₦'000	₦'000	₦'000
Loan Advances	1,500						1,500
Debtors	214	642	642	642	642	642	3,421
	1,714	642	642	642	642	642	4,921
Payments:							
Capital Exp	983						983
Trade Creditors	257	342	342	342	342	342	1,967
Loan Rept			100	100,	100	50	350
Sal and Wages	44	44	44	44	44	44	266
Overhds/Exp	218	218	218	218	218	218	1,307
	1,502	**604**	**704**	**704**	**704**	**654**	4,874
Difference	212	37	-63	-63	-63	-13	47
Balance B/F	0	212	249	186	123	60	0
Balance C/F	212	249	186	123	60	47	47
Current Assets							
Debtors	0	214	428	641	855	1,069	1,283
Stock	26	499	499	499	499	499	499
Cash/Bank	0	212	249	186	123	60	47
	26	**924**	**1,175**	**1,326**	**1,477**	**1,628**	**1,829**
Account Payable	0	-158	-231	-303	-376	-449	-521
Working Capital	**26**	**766**	**945**	**1,023**	**1,101**	**1,180**	**1,308**

5.4 CONCLUSIONS /RECOMMENDATION

The financial projection incorporates the Profit and Loss Account, cash flow and working capital analysis for six months operation. The initial capital funding of N2.6m although not shown in the projections is made up of:

1. Product processing machines
2. Factory House construction
3. CAC Incorporation and other legal expenses

The firm may require revolving funds of about N1.5 million naira to commence operations and operate for six months run. This is for:

1. The construction of the factory structures,
2. The Heat rooms
3. The drying platforms
4. The production machines
5. A utility vehicle,
6. Stock of raw products
7. Salaries and wages for six months.
8. Overheads and administration expenses

This project is viable. At 60% projected operations, the working capital keeps increasing comprising of Debtors. Debtors are cash convertible current asset.

By the end of the six months, the operations should commence repayment funding if financed with a loan. Therefore, within one year the factory operations would have been self sustaining.

5.5 FLASHCARD - CASH FLOW AND FINANCIAL MANAGEMENT

1. **Q:** What is Cash Flow?
   A: Movement of money into and out of a business.
2. **Q:** What is Gross Profit?
   A: Sales revenue minus direct production costs.
3. **Q:** What is Working Capital?
   A: Funds used for daily business operations.
4. **Q:** What are Current Assets?
   A:
   - Cash
   - Inventory
   - Debtors
   - Bank balances
5. **Q:** What is Break-even Point?
   A: The point where total revenue equals total costs.
6. **Q:** Why is Cash Flow Forecasting important?
   A: To support planning, financial control, and investment decisions.

COMPREHENSIVE REVIEW

7. **Q:** What is Agro-Industrialization?
   A: The transformation of agricultural products into industrial inputs and consumer goods.
8. **Q:** What is Food Preservation?
   A: Techniques used to prevent spoilage and increase product shelf life.

9. **Q:** What is a Value Chain?

   A: The complete sequence of activities that add value from production to consumption.

10. **Q:** What is Entrepreneurship?

    A: The ability to identify opportunities and organize resources to create value and profit.

11. **Q:** What is Sustainable Agriculture?

    A: Agricultural practices that meet current needs without compromising future generations.

Multiple Choice Questions

1. Cash flow projection is primarily used for:

   a) Tax avoidance

   b) Profit planning

   c) Employee recruitment

   d) Licensing

   Answer: b

2. Planned operation starts at:

   a) 20%

   b) 40%

   c) 60%

   d) 100%

   Answer: c

3. Input-output ratio is:

   a) 1:1

   b) 2:1

   c) 3:1

d) 4:1

Answer: b

4. Gross profit becomes positive from:

a) Month 1

b) Month 2

c) Month 4

d) Month 6

Answer: b

5. Debtors are classified as: a) Fixed assets b) Current assets c) Long-term liabilities d) Capital assets

Answer: b

Numerical Problems

Question 1

If:

- Raw product cost = ₦30,000/ton
- Dried product selling price = ₦150,000/ton

Calculate:

a. Gross margin per ton

b. Gross profit percentage

Question 2

A factory purchases 20 tons of cassava roots.

Input-output ratio = 2:1

Calculate:

a. Expected dried output

b. Expected sales revenue

Question 3

A factory operates at:

- 60% capacity in Year 1

- 70% capacity in Year 2
- 80% capacity in Year 3

Estimate annual production growth.

Essay Questions

1. Explain the importance of cash flow forecasting in agro-processing businesses.
2. Discuss factors affecting profitability in Dry Harvest enterprises.
3. Evaluate the financial viability of a cassava flour processing business.

5.6 **COMPREHENSIVE FINAL PRACTICE EXAMINATION**

SECTION A (20 Marks)

20 Multiple Choice Questions drawn from all chapters.

SECTION B (30 Marks)

Answer any three:

1. Explain the Dry Harvest entrepreneurship model.
2. Discuss capacity management in agro processing.
3. Compare artificial and natural drying methods.
4. Describe a complete cassava flour production process.
5. Explain market-chain development in dry-harvest industries.

5.7 SECTION C: CASE STUDY (50 Marks)

You have secured ₦10 million to establish a Dry Harvest Processing Factory producing:

- Cassava Flour
- Yam Flour
- Plantain Flour
- Pellets

Prepare:

1. Business Model (10 Marks)
2. Capacity Plan (10 Marks)
3. Production Process Design (10 Marks)
4. Marketing & Distribution Strategy (10 Marks)
5. Cash Flow Forecast and ROI Analysis (10 Marks)

Total = 100 Marks

5.8 **ASSESSMENT SCORING GUIDELINES**

Quiz Assessment (20 Marks)

Score	Grade	Performance
18-20	A	Excellent
16-17	B+	Very Good
14-15	B	Good
12-13	C	Satisfactory
10-11	D	Fair
Below 10	F	Fail

Short Answer Assessment (20 Marks)

Marking Scheme

Requirement	Marks
Correct definition	2
Explanation	2
Relevant example	1

Total = 5 Marks per Question

4 Questions × 5 Marks = 20 Marks

Essay Assessment (20 Marks)

Marking Rubric

Criteria	Marks
Understanding of topic	5
Depth of analysis	5
Practical application	5
Organization and clarity	3

Conclusion	2

Total = 20 Marks

Case Study Assessment (40 Marks)

Rubric

Section	Marks
Problem identification	5
Data analysis	10
Application of concepts	10
Recommendations	10
Presentation quality	5

Total = 40 Marks

5.9 FINAL CERTIFICATION EXAMINATION MARKING GUIDE

Section	Marks
Multiple Choice Questions	20
Short Answer Questions	20
Essay Questions	20
Case Study Analysis	40

TOTAL = 100 MARKS

5.10 CERTIFICATION CLASSIFICATION

Score (%)	Grade	Classification
85-100	A	Distinction
75-84	B+	Upper Credit
65-74	B	Credit
50-64	C	Pass
40-49	D	Conditional Pass
Below 40	F	Fail

Facilitator Recommendation

For certification programmes, use:

- **30% Continuous Assessment**
  - Chapter Quizzes (10%)
  - Flashcard Tests (10%)
  - Assignments (10%)
- **70% Final Examination**
  - MCQs (20%)
  - Essays (20%)
  - Practical Case Study (30%)

5.11 ADDITIONAL CASE STUDIES

The following case studies may be used for assignments, practical assessments, class discussions, project work, certification examinations, and entrepreneurial coaching.

CASE STUDY 1: CASSAVA FLOUR PROCESSING ENTERPRISE

A cooperative society in Oyo State has secured ₦15 million to establish a cassava flour processing plant.

The cooperative intends to:

- Purchase cassava roots from local farmers
- Produce High-Quality Cassava Flour (HQCF)
- Supply bakeries and food industries
- Export surplus production to neighboring countries

Production Data

Item	Quantity
Cassava Purchase	500 Tons
Cost per Ton	₦35,000
Processing Cost per Ton	₦12,000
Expected Conversion Ratio	2:1
Selling Price of HQCF	₦180,000/Ton

Questions

1. Calculate the total raw material cost.
2. Estimate total flour output.
3. Determine total sales revenue.
4. Calculate gross profit.
5. Recommend a marketing strategy.

Marks Allocation (20 Marks)

- Financial Calculations = 10 Marks
- Business Analysis = 5 Marks
- Recommendations = 5 Marks

CASE STUDY 2: PLANTAIN FLOUR BUSINESS START-UP

A graduate entrepreneur wants to establish a plantain flour business targeting:

- Diabetic patients
- Health-conscious consumers
- Hospitals
- Fitness centers

Available startup capital = ₦5 million.

Problem Statement

The entrepreneur must decide whether to:

Option A: Manual drying operation

Option B: Mechanical drying operation

Information

Variable	Manual Drying	Mechanical Drying
Startup Cost	₦2m	₦4m
Daily Capacity	200kg	800kg
Product Quality	Moderate	High
Drying Duration	3 Days	8 Hours

Questions

1. Compare both options.
2. Identify risks.
3. Recommend the better alternative.
4. Explain how quality affects marketability.

Assessment Focus

- Decision Making
- Cost-Benefit Analysis
- Capacity Planning

CASE STUDY 3: COMMUNITY DRYING CENTER

A rural community loses approximately 40% of harvested agricultural produce annually because of poor preservation facilities. Products include:

- Cassava
- Yam
- Banana
- Plantain

A donor agency proposes a Community Dry Harvest Centre. Objectives

- Reduce post-harvest losses
- Create employment
- Improve food security
- Increase rural income

Questions

1. Identify the major causes of losses.
2. Design a suitable drying system.
3. Estimate community benefits.
4. Recommend management structures.
5. Develop sustainability measures.

Marks

Section	Marks
Problem Analysis	5
Technical Solution	5
Economic Benefits	5
Sustainability Plan	5

Total = 20 Marks

CASE STUDY 4: AGRO-INDUSTRIAL VALUE CHAIN DEVELOPMENT

A state government plans an Agro-Industrial Hub integrating:

- Farms
- Processing plants
- Warehouses
- Transport systems
- Export terminals

Investment budget: ₦1.2 Billion

Challenge: Develop a value-chain model that maximizes:

- Job creation
- Local sourcing
- Export revenue
- Food security

Questions

1. Draw a value-chain framework.
2. Identify critical stakeholders.
3. Describe revenue sources.
4. Explain expected social impact.
5. Recommend performance indicators.

Practical Deliverable **-** Students prepare:

- Value-chain map
- Capacity plan
- Performance dashboard

CASE STUDY 5: CASSAVA CHIP EXPORT PROJECT

A company receives an export order for 1,500 tons of cassava chips.

Requirements:

Requirement	Value
Moisture Content	Below 12%
Delivery Time	60 Days
Packaging	Export Standard
Capacity Available	1,000 Tons

Questions

1. Determine the capacity gap.
2. Propose solutions.
3. Assess operational risks.
4. Develop an export logistics plan.
5. Recommend quality controls.

Expected Learning Outcomes

- Export Readiness
- Supply Chain Planning
- Quality Assurance
- Capacity Management

CASE STUDY 6: AGRICULTURAL BUSINESS RESCUE PROJECT

A drying factory has recorded losses for three consecutive years.

Findings

- Equipment downtime = 35%

- Inventory losses = 18%
- Capacity utilization = 40%
- Product rejection = 12%

Assignment

Act as a Management Consultant.

Questions

1. Identify root causes.
2. Conduct SWOT analysis.
3. Recommend improvement initiatives.
4. Estimate impact of recommendations.
5. Develop an implementation roadmap.

Marking Guide

Criteria	Marks
Root Cause Analysis	10

5.12 ADDITIONAL REAL-WORLD CASE STUDIES

These case studies are designed to strengthen practical knowledge, business decision-making, export readiness, financial management capability, and risk assessment skills in Dry-Harvest Agricultural Entrepreneurship.

SECTION A: REAL-WORLD AGRIBUSINESS CASE STUDIES

Case Study 10: High-Quality Cassava Flour (HQCF) Supply to Bakeries

A cassava processing company has signed agreements with 20 bakeries that intend to replace 20% of wheat flour with High-Quality Cassava Flour (HQCF).

Business Data

- Monthly bakery demand: 300 tons
- HQCF selling price: ₦220,000/ton
- Production cost: ₦150,000/ton
- Current capacity: 250 tons/month

Challenges

- Production shortfall
- Quality consistency
- Delivery scheduling
- Competition from wheat flour suppliers

Tasks

1. Calculate monthly revenue potential.
2. Determine capacity gap.
3. Recommend capacity expansion strategies.
4. Develop a customer retention plan.
5. Identify operational risks.

Learning Areas

- Capacity Planning
- Contract Manufacturing
- Market Development
- Customer Relationship Management

CASE STUDY 11: PLANTAIN FLOUR HEALTH MARKET EXPANSION

A plantain flour company is targeting:

- Diabetic consumers
- Hospitals
- Nutrition clinics
- Health stores

Recent Performance

Indicator	Value
Monthly Production	30 Tons
Monthly Sales	25 Tons
Market Growth Rate	18%
Product Rejection	2%

Questions

1. Analyze market opportunities.
2. Identify customer segments.
3. Develop a growth strategy.
4. Design a promotional campaign.
5. Estimate future market demand.

Case Study 12: Youth Agro-Processing Incubation Centre

A state government establishes an incubation center for:

- Cassava processors
- Yam flour processors
- Plantain flour processors
- Pellet producers

Target beneficiaries: 500 unemployed youths.

Assignment

Prepare:

- Training Plan
- Business Incubation Framework
- Funding Model
- Revenue Strategy
- Monitoring Framework

Marks: 30

SECTION B: EXPORT LOGISTICS CASE STUDIES

Case Study 13: Exporting Cassava Flour to the United Kingdom

An exporter receives an order for 200 tons of High-Quality Cassava Flour from a food manufacturing company in the United Kingdom.

Export Requirements

Requirement	Specification
Product	HQCF
Quantity	200 Tons
Delivery Period	45 Days
Moisture Content	Below 12%
Packaging	25kg Export Bags
Shipping Port	Lagos Port

Challenges

- Documentation preparation
- Quality certification
- Container booking
- Exchange-rate fluctuations

Questions

1. Identify export documents required.
2. Develop a shipping timetable.
3. Recommend quality assurance controls.
4. Estimate logistics risks.
5. Design a contingency plan.

Expected Documents

- Commercial Invoice
- Packing List
- Bill of Lading
- Certificate of Origin
- Phytosanitary Certificate
- Export Permit
- Insurance Certificate

Learning Outcomes

- Export Readiness
- Logistics Planning
- International Trade Documentation

Case Study 14: Container Delay at Seaport

A company exports 50 containers of cassava chips. Two weeks before vessel departure:

- Port congestion increases
- Shipping schedules are delayed
- Storage fees begin accumulating

Financial Data

Item	Cost
Daily Demurrage	₦150,000
Daily Storage Fee	₦80,000
Number of Delayed Days	12

Questions

1. Calculate additional logistics costs.
2. Assess financial impact.
3. Recommend mitigation strategies.
4. Develop a crisis management plan.

5. Identify preventive measures.

Practical Exercise

Prepare:

- Delay Report
- Executive Risk Memo
- Corrective Action Plan

Case Study 15: Export Rejection Due to Quality Issues

A shipment of plantain flour is rejected by a foreign buyer because moisture levels exceed agreed standards.

Losses

Item	Value
Production Cost	₦18 Million
Logistics Cost	₦5 Million
Product Recall Cost	₦2 Million
Contract Penalty	₦3 Million

Questions

1. Determine total financial loss.
2. Conduct root-cause analysis.
3. Recommend quality improvements.
4. Design an export quality-control framework.
5. Develop preventive measures.

Assessment Areas

- Quality Control
- Supply Chain Management
- Export Compliance

SECTION C: FINANCIAL RISK MANAGEMENT CASE STUDIES

Case Study 16: Rising Raw Material Prices

A cassava flour company experiences rapid increases in cassava root prices.

Cost Data

Year	Cassava Cost/Ton
Year 1	₦30,000
Year 2	₦40,000
Year 3	₦55,000

Current Situation

- Selling price remains fixed.
- Profit margins are declining.

Questions

1. Calculate percentage increases.
2. Assess financial risks.
3. Recommend pricing strategies.
4. Develop supplier-risk management plans.
5. Suggest cost-control measures.

Learning Objective: Understanding commodity-price risk.

Case Study 17: Foreign Exchange (FX) Risk

An exporter receives payment in US Dollars. **Contract Value:** USD 250,000

Exchange Rates

Date	Exchange Rate
Contract Date	₦1,300/$
Shipment Date	₦1,500/$
Payment Date	₦1,200/$

Questions

1. Calculate expected naira revenue at each date.
2. Determine exchange-rate loss/gain.
3. Identify FX risks.
4. Recommend hedging strategies.
5. Develop a foreign-currency policy.

Learning Areas

- Currency Risk Management
- International Finance
- Export Revenue Protection

Case Study 18: Equipment Breakdown Risk

A drying facility's mechanical dryer fails during peak production season.

Operational Impact

Item	Value
Production Loss	40 Tons
Sales Value Lost	₦8 Million
Repair Cost	₦1.5 Million
Downtime	15 Days

Questions

1. Calculate total business impact.
2. Explain operational risks.
3. Recommend preventive maintenance strategies.
4. Develop a business continuity plan.
5. Assess insurance options.

Learning Areas

- Operational Risk
- Business Continuity
- Asset Management

Case Study 19: Credit Risk and Customer Default

A distributor purchases goods on credit.

Outstanding receivable: ₦12 Million
The customer fails to pay after 120 days.

Questions

1. Assess the financial impact.
2. Recommend debt recovery strategies.
3. Design a credit policy.
4. Identify warning signs of bad debts.
5. Recommend controls to reduce future defaults.

Risk Management Focus

- Credit Management
- Cash Flow Protection
- Working Capital Control

Case Study 20: Agro-Processing Investment Risk Analysis

An investor plans to establish a ₦100 Million agro-processing facility.

Investment Distribution

Item	Amount
Land	₦15 Million
Buildings	₦25 Million
Machinery	₦40 Million
Working Capital	₦20 Million

Potential Risks

- Market Demand Risk
- Inflation Risk
- Technology Risk
- Regulatory Risk
- Supply Chain Risk
- Environmental Risk

Questions

1. Prepare a risk register.
2. Conduct SWOT analysis.
3. Rank risks by severity.
4. Recommend mitigation actions.
5. Develop a risk monitoring framework.

Expected Deliverable

Professional Enterprise Risk Management (ERM) Report.

ADVANCED CAPSTONE CASE STUDY: EXPORT-ORIENTED DRY-HARVEST INDUSTRIAL COMPLEX

Scenario

You have been appointed Managing Director of a newly established export-oriented Dry-Harvest Agro-Industrial Complex.

Project Scope

- Cassava Flour Production
- Plantain Flour Production
- Yam Flour Production
- Pellet Manufacturing
- Export Logistics Centre
- Warehouse and Distribution Hub

Investment Budget: ₦5 Billion

Export Markets

- United Kingdom
- Netherlands
- Germany
- Ghana
- South Africa

Assignment

Prepare a comprehensive professional report covering:

Part A: Strategic Business Plan (20 Marks)

- Market Analysis
- Business Objectives
- Growth Strategy

Part B: Export Logistics Plan (20 Marks)

- Export Documentation
- Shipping Model
- Port Operations
- Customs Compliance

Part C: Financial Risk Assessment (20 Marks)

- FX Risk
- Credit Risk

- Liquidity Risk
- Inflation Risk

Part D: Operational Risk Management (20 Marks)

- Production Risks
- Quality Risks
- Supply Risks
- Technology Risks

Part E: Financial Viability Analysis (20 Marks)

- ROI
- NPV
- Payback Period
- Sensitivity Analysis

Total = 100 Marks

Professional Grading Guide

Score	Classification
85-100	Distinction
75-84	Upper Credit
65-74	Credit
50-64	Pass
Below 50	Fail

GLOSSARY OF TERMS

A Quick Guide to Dry-Harvest Agricultural Entrepreneurship

A

Agri-Business A business engaged in agricultural production, processing, distribution, marketing, or support services.

Agro-Industrialization The transformation of agricultural raw materials into value-added products through processing and manufacturing.

Agricultural Entrepreneurship The process of identifying, developing, and managing agricultural business opportunities for profit and sustainable development.

Agricultural Value Chain The series of activities involved in moving agricultural products from the farm to the final consumer.

Asset A resource owned by a business that has economic value and can generate future benefits.

B

Balance Sheet A financial statement showing assets, liabilities, and owner's equity at a particular date.

Branding The process of creating a unique identity, image, and reputation for a product or business.

Break-Even Point The level at which total revenue equals total costs, resulting in neither profit nor loss.

Business Continuity Plan A documented strategy for maintaining operations during disruptions or emergencies.

Business Plan A formal document outlining business objectives, strategies, financial forecasts, and operational plans.

C

Capacity Management The planning, monitoring, and optimization of resources to achieve desired production levels.

Capacity Utilization The percentage of available production capacity that is being used.

Capital Resources used to establish and operate a business.

Cash Flow The movement of money into and out of a business.

Commodity A basic agricultural product that can be bought or sold.

Competitive Advantage A factor that enables a business to outperform competitors.

Credit Risk The risk that a customer or borrower may fail to fulfil financial obligations.

Current Assets expected to be converted into cash within one year.

D

Debtor A customer who owes money to a business.

Depreciation The gradual reduction in the value of an asset due to use or age.

Distribution Channel The route through which products move from producer to consumer.

Dry Harvest Management The coordinated management of drying, storage, processing, packaging, distribution, and marketing of agricultural products.

Drying The removal of moisture from agricultural products to increase shelf life and maintain quality.

Drying Production System (DPS) A production system designed to preserve and process agricultural products through moisture reduction.

E

Enterprise An organized business activity established to generate value and profit.

Entrepreneur An individual who identifies opportunities and organizes resources to establish and manage a business.

Enterprise Risk Management (ERM) A structured approach to identifying, assessing, and controlling risks across an organization.

Export The sale of goods and services to buyers in foreign countries.

Export Logistics The planning and management of transportation, storage, documentation, and delivery of products to international markets.

F

Farmer Cooperative An association of farmers who jointly engage in production, processing, marketing, or purchasing activities.

Financial Capital Money and financial resources used to establish and operate a business.

Financial Risk The possibility of financial loss arising from market changes, credit issues, inflation, or operational failures.

Food Preservation Methods used to reduce spoilage and extend the useful life of food products.

Foreign Exchange (FX) Risk The risk of financial loss resulting from fluctuations in currency exchange rates.

G

Gari A roasted granular food product processed from fermented cassava.

Good Manufacturing Practice (GMP) Procedures and standards that ensure products are consistently produced and controlled according to quality requirements.

Gross Profit Revenue remaining after deducting direct production costs.

H

Human Capital The skills, knowledge, experience, and competence possessed by individuals.

High-Quality Cassava Flour (HQCF) A refined cassava flour produced to meet industrial and food-processing standards.

I

Incubation Centre A facility established to support entrepreneurs through training, mentoring, and business development services.

Industrial Chain The interconnected sequence of activities linking production, processing, storage, distribution, and marketing.

Inflation Risk The risk that rising prices will reduce profitability and purchasing power.

Inventory Raw materials, work-in-progress items, and finished goods held for future use or sale.

K

Key Performance Indicator (KPI) A measurable value used to evaluate business performance and achievement of objectives.

L

Liquidity The ability of a business to meet its short-term financial obligations.

Logistics The management of transportation, handling, storage, and movement of goods.

M

Market Segmentation The process of dividing a market into groups of customers with similar characteristics or needs.

Marketing Chain The flow of products from producers through intermediaries to final consumers.

Mechanical Drying The use of machines and controlled heat to reduce moisture content in products.

Microfinance Financial services provided to low-income individuals and small business operators.

Moisture Content The percentage of water present in an agricultural product.

N

Net Profit The amount remaining after all business expenses, taxes, and costs have been deducted from revenue.

NPV (Net Present Value) A financial evaluation tool used to determine the current value of future cash flows.

O

Operational Risk The risk of loss resulting from failed processes, equipment breakdowns, human error, or external events.

P

Packaging The process of enclosing products for protection, transportation, storage, and marketing.

Payback Period The time required to recover the original investment from business profits.

Pellet A compressed feed product made from agricultural materials and by-products.

Post-Harvest Loss The reduction in quantity or quality of agricultural products after harvest.

Processing The conversion of raw agricultural materials into finished or semi-finished products.

Profitability The ability of a business to generate profits relative to costs and investments.

Q

Quality Assurance (QA) Systematic activities designed to ensure products consistently meet required standards.

Quality Control (QC) Inspection and testing activities used to verify product quality.

R

Raw Material The basic agricultural product used in production processes.

Return on Investment (ROI) A measure of profitability expressed as:

ROI = Net Profit ÷ Investment × 100

Risk Register A document used to record identified risks, their likelihood, impact, and mitigation actions.

S

Sensitivity Analysis A technique used to evaluate how changes in key variables affect business outcomes.

Shelf Life The period during which a product remains safe and suitable for use.

Social Capital The value derived from relationships, networks, trust, and cooperation among individuals and organizations.

Stakeholder Any person, group, or organization affected by or interested in a business activity.

Storage The safe keeping of products before use, processing, or sale.

Supply Chain The network involved in producing and delivering products to consumers.

Sustainability Practices that balance economic growth, environmental protection, and social wellbeing.

SWOT Analysis A strategic assessment tool used to identify Strengths, Weaknesses, Opportunities, and Threats.

T

Technology Transfer The process of moving knowledge, skills, or innovations from research institutions to commercial application.

Throughput The quantity of products processed within a specific time.

V

Value Addition The enhancement of products through processing, packaging, branding, or quality improvement.

Value Chain All activities involved in creating, processing, marketing, and delivering a product to the customer.

Value Proposition The unique benefits offered by a product or service to customers.

W

Warehouse A facility used for storing raw materials, intermediate products, or finished goods.

Working Capital Funds available for day-to-day business operations.

Y

Yield The quantity of finished product obtained from a given quantity of raw materials.

Recommended Acronyms

Acronym	Meaning
DPS	Drying Production System
ERM	Enterprise Risk Management
FX	Foreign Exchange
GMP	Good Manufacturing Practice
HQCF	High-Quality Cassava Flour
KPI	Key Performance Indicator
NPV	Net Present Value
QA	Quality Assurance
QC	Quality Control
ROI	Return on Investment
SOP	Standard Operating Procedure
SWOT	Strengths, Weaknesses, Opportunities and Threats

CHAPTER REVIEWS

Chapter 1: Dry-Harvest Entrepreneurship

Chapter One introduces readers to the concept of Dry-Harvest Entrepreneurship and establishes the foundation upon which the rest of the book is built. The chapter effectively explains the relationship between agricultural production, entrepreneurship, and value creation. It highlights the importance of human capital, social capital, financial resources, and business planning in establishing sustainable agribusiness ventures.

A major strength of this chapter is its focus on entrepreneurship as a vehicle for economic empowerment rather than merely income generation. The discussion on self-employment, enterprise creation, and microfinance options provides aspiring agripreneurs with practical insights into launching and managing agricultural businesses.

Strengths

- Clear introduction to agribusiness entrepreneurship.
- Practical explanation of business planning.
- Strong linkage between agriculture and wealth creation.
- Useful for youth entrepreneurship programmes.

Rating: ★★★★★ (4.8/5)

Chapter 2: Capacity Management Concept

This chapter examines the strategic importance of capacity management in agro-processing enterprises. It provides readers with an understanding of production capacity,

resource optimization, profitability analysis, return on investment, and industrial-chain development.

The author's approach is commendable because it links operational planning directly to business performance. Readers gain an appreciation for how production systems, human resources, infrastructure, and technology influence productivity and profitability.

The chapter is especially beneficial for managers, cooperative leaders, investment analysts, and project developers involved in agro-industrial ventures.

Strengths

- Strong business planning perspective.
- Practical ROI and payback concepts.
- Effective illustration of industrial-chain development.
- Useful for project feasibility assessment.

Rating: ★★★★★ (4.7/5)

Chapter 3: Technology Development

Chapter Three provides a comprehensive overview of drying technologies, processing methods, preservation systems, and agro-processing innovations. It serves as the technical core of the book and demonstrates how agricultural products can be transformed into higher-value commodities.

Particularly noteworthy is the discussion on drying systems, cassava flour production, gari production, pellet manufacturing, and moisture management. Complex technical processes are explained in a simple and accessible manner, making the content suitable for both beginners and practitioners.

The chapter succeeds in illustrating how technology adoption can enhance efficiency, reduce post-harvest losses, and improve product quality.

Strengths

- Excellent practical orientation.
- Strong focus on post-harvest management.
- Clear explanation of processing technologies.
- Promotes industrial-scale thinking.

Rating: ★★★★★ (4.9/5)

Chapter 4: Sales and Distribution Management

Chapter Four addresses one of the most overlooked areas of agricultural entrepreneurship: market access. The chapter explains how products move from farms and processing facilities to local and international markets through effective distribution and marketing systems.

The author emphasizes value addition, branding, packaging, distribution planning, customer targeting, and market segmentation. The discussion helps readers understand that profitability depends not only on production but also on successful market penetration.

This chapter is particularly useful for entrepreneurs seeking to expand beyond local markets and enter national or export channels.

Strengths

- Practical marketing frameworks.
- Strong emphasis on value-chain integration.
- Excellent introduction to distribution management.
- Relevant export market considerations.

Rating: ★★★★★ (4.8/5)

Chapter 5: Hypothetical Cash Flow Projection

The final chapter deals with the financial realities of agribusiness operations. It introduces readers to cash flow forecasting, investment planning, profitability analysis, working capital management, and financial sustainability.

One of the greatest strengths of this chapter is its ability to simplify financial concepts that are often intimidating to entrepreneurs. Through practical examples and projections, readers learn how financial planning supports business continuity and investment decision-making.

The chapter provides an excellent foundation for entrepreneurs seeking funding from banks, investors, development agencies, or cooperative finance institutions.

Strengths

- Practical financial modeling.
- Useful for investment readiness.
- Strong cash flow management concepts.
- Supports informed decision-making.

Rating: ★★★★★ (4.8/5)

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A QUICK GUIDE TO

DRY-HARVEST AGRICULTURAL ENTREPRENEURSHIP



THE MISSING VALUE

I harvested **10 tons** but earned almost nothing!

I harvested **value**, not just cassava!

DRY HARVEST ENTREPRENEUR

VALUE ADDITION CREATES OPPORTUNITIES

DRY HARVEST PROCESSING

- HIGHER INCOME
- JOB CREATION
- FOOD SECURITY
- EXPORT MARKET

FRESH CASSAVA ROTTING (NO VALUE ADDED)

CASSAVA FLOUR — Nutritious Versatile Profitable

GARI — Local Taste Global Market

PELLETS — Animal Feed More Value

EXPORT CHIPS — Quality for the World

VALUE ADDITION CREATES WEALTH

FROM FARM TO PROSPERITY

ICIME PUBLISHING

Enterprise • Innovation • Impact

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